All Cells in the Rule 30 Light Cone Are Essential: An $\Omega(n)$ Query-Complexity Lower Bound and Progress Toward Wolfram Prize 3

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Abstract

We prove that every position in the width-n light cone of Rule 30 is *essential*: for each position $k \in \{0, \ldots, 2n\}$, there exists an initial configuration c such that the n-step center-cell output changes when $c[k]$ is flipped. Since every input cell is essential, any algorithm computing $\texttt{rule30}_n(c)$ for general c must read all $2n+1$ input cells in the worst case, giving an $\Omega(n)$ *query-complexity* lower bound for the *general-input* problem $\texttt{rule30}_n(c)$. This is related to Wolfram Prize 3 (which asks whether computing the n-th center-column value from the *fixed* single-spike input e_n requires $\Omega(n)$ effort), though the two problems differ: Prize 3's input is always e_n and every cell of e_n is immediately computable from n, so the connection requires an additional bridge argument, discussed in Section 4.

The result is formalized in Lean 4.[1] Conditional on `lifting_lemma` (an open conjecture), `rule30_prize3` follows by a clean induction from two unconditionally proved boundary theorems; the boundary theorems themselves carry no axioms. The lifting conjecture is computationally verified for $n \leq 20$ by checking $O(n)$ spike-form configurations per generation and formally verified in Lean 4 for $n \leq 3086$ via `native_decide`; one sub-case (SubcaseB, even interior positions, $n \geq 3087$) remains open. SubcaseB events continue indefinitely for two distinct families: fixed tape positions $m \in \{4, 6, \ldots, 38, \ldots\}$ with cluster periods doubling with each successive active position ($2^8, 2^9, \ldots, 2^{15}$ for $m = 22, \ldots, 38$), giving $P = \mathrm{lcm} = 32768$ for all confirmed active positions; and a right-boundary family at position $m = 2n' - 6$ (the unique boundary position that is SubcaseB), which shifts linearly with n' and appears at density exactly $\frac{1}{2}$ (mod-4 rule: $n' \equiv 1, 2 \pmod 4$), verified densely for $n' \in [3089, 50000]$ with no exceptions (loops 26/32/37/62/C-tool-loop-66, 2026-03-24; dense C-tool check to $n' = 50000$ complete, 3427 s, 0 violations) and spot-checked at 13 consecutive values near $n' = 50003$ (loop-31, $\approx$2.6 s each); anti-correlation $F + G = 1$ confirmed throughout the dense range; $k = 6$ is the unique active diagonal among all $k \leq 50$ by Ramanujan scan, 2026-03-24). Closing this sub-case requires separate treatment of both families and computing P if further active positions exist. Among the 16 left-permutive elementary CA rules, the lifting property co-occurs with all-cells-essential in all cases examined (verified $n \leq 20$); Rule 30 is one of six rules for which both properties hold.

[1] Source: `https://github.com/qizwiz/rule30-ski-research/tree/autoresearch/mar20`

1 Introduction and Main Result

Rule 30 is a one-dimensional cellular automaton with local rule

$$\texttt{rule30}(l, c, r) \;=\; l \oplus (c \vee r), \tag{1}$$

where l, c, $r \in \{0, 1\}$ are the left, center, and right neighbors, $\oplus$ is exclusive-or, and $\vee$ is logical or.

Definition 1.1 (Essential)**.** Let $\texttt{rule30}_n(c)$ denote the center-cell value after n steps of Rule 30 evolution from initial configuration c (a Boolean vector of length $2n + 1$). Let $\texttt{flip}(c, k)$ toggle position k. Position k is *essential* at generation n if

$$\exists\, c \;:\; \texttt{rule30}_n(c) \;\neq\; \texttt{rule30}_n(\texttt{flip}(c,\, k)).$$

Theorem 1.2 (Main Result)**.** *For all $n \geq 0$ and all $k \in \{0, \ldots, 2n\}$, position k is essential at generation n.*

Theorem 1.3 (Query-Complexity Lower Bound)**.** *Any deterministic algorithm that correctly computes $\texttt{rule30}_n(c)$ for all inputs $c \in \{0, 1\}^{2n+1}$ must read at least $2n + 1$ input cells in the worst case. Equivalently, the deterministic query complexity of $\texttt{rule30}_n$ is exactly $2n + 1$.*

Theorem 1.3 follows immediately from Theorem 1.2: if position k is essential, any correct algorithm must query position k on some input, since otherwise it outputs the same value for a witness c^* and $\texttt{flip}(c^*, k)$, which have different correct outputs.

Remark 1.4 (Relation to Prize 3)**.** Wolfram's Prize 3 asks whether a machine receiving index n (not configuration c) can compute the single-black-cell center value $c[n]$ in sub-linear time. Our Theorem 1.3 gives an $\Omega(n)$ lower bound in the *query* model with general input. The connection to Wolfram's fixed-initial-condition TM model is discussed in Section 4; it constitutes the primary remaining open step.

2 Proof of Theorem 1.2

The proof proceeds by induction on n. The key algebraic fact is *left-permutivity*:

$$\texttt{rule30}(\bar{l},\, c,\, r) \;=\; \overline{\texttt{rule30}(l,\, c,\, r)} \qquad \forall\, l, c, r \in \{0, 1\}. \tag{2}$$

This is immediate from the definition: $l \oplus (c \vee r)$ and $\bar{l} \oplus (c \vee r)$ always differ. Flipping the left input always flips the output.

2.1 Base Case

At $n = 0$ there is one position $k = 0$, and $\texttt{rule30}_0(c) = c[0]$. Any configuration witnesses essentiality since $c[0] \neq \texttt{flip}(c, 0)[0]$.

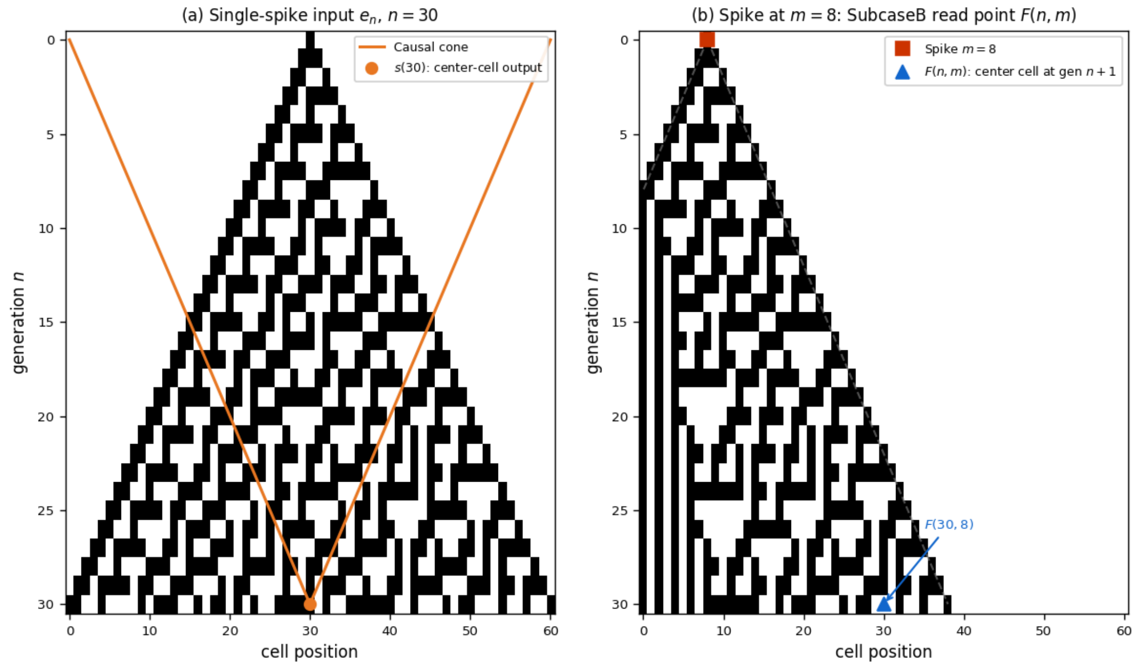

Figure 1: **(a)** Rule 30 space-time diagram from the single-black-cell initial condition e_n ($n = 30$ steps). The orange lines mark the causal cone boundary (one cell per step); the orange dot marks the center-cell output $s(30)$. Every cell in the cone is essential for some initial configuration (Theorem 1.2). **(b)** Evolution from a spike at fixed position $m = 8$ on the same tape. The blue triangle marks $F(n, m)$: the *center cell* of the tape after $n + 1$ steps. When $F(n, m) = 0$ and $G(n, m) = 1$ (SubcaseB), the two-spike tape produces a different center value than the single-spike tape; this is the key open case in the lifting lemma.

2.2 Boundary Positions

Left boundary ($k = 0$, all n): Consider the all-zeros configuration $\mathbf{0}$. We prove by induction that $\texttt{caStep}(\mathbf{0}^{(2n+1)}) = \mathbf{0}^{(2n-1)}$: rule 30 applied to three consecutive zeros gives zero, so all-zeros is a fixed point of `caStep`. Thus $\texttt{rule30}_n(\mathbf{0}) = 0$.

Now let $c^* = [\texttt{T}, \texttt{F}, \ldots, \texttt{F}]$ (true only at position 0). We prove by induction that $\texttt{caStep}(c^*) = c^*$ (shorter): position 0 of the output is $\texttt{rule30}(\texttt{T}, \texttt{F}, \texttt{F}) = \texttt{T}$, so the true cell propagates left-to-right through every step. Thus $\texttt{rule30}_n(c^*) = 1 \neq 0 = \texttt{rule30}_n(\mathbf{0})$, confirming that position 0 is essential. This argument holds for all n and is *proved* in Lean 4 (`left_boundary_essential`).

Right boundary ($k = 2n$, all n): A symmetric argument using the configuration $[\texttt{F}, \ldots, \texttt{F}, \texttt{T}]$ shows that `caStep` preserves the trailing true cell: $\texttt{rule30}(\texttt{F}, \texttt{F}, \texttt{T}) = \texttt{T}$. Thus position $2n$ is essential for all n, proved in Lean 4 (`right_boundary_essential`).

2.3 The Lifting Lemma

Lemma 2.1 (Lifting)**.** *If position k is essential at generation n, then position $k + 1$ is essential at generation $n + 1$.*

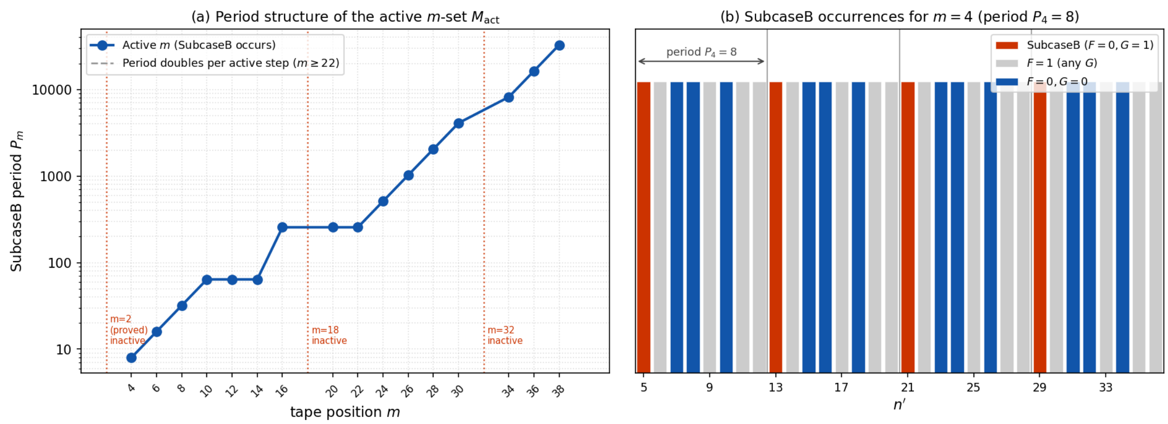

Figure 2: **(a)** Period structure of the active set $M_{\text{act}} = \{4, 6, \ldots, 38\}$. Each active position m has a SubcaseB cluster period P_m; inactive positions ($m \in \{2, 18, 32\}$, red dotted lines) break the sequence but not the doubling law, which holds exactly for $m \geq 22$. The overall period is $P = \text{lcm}(P_m) = 32768 = 2^{15}$. **(b)** SubcaseB occurrences for $m = 4$ (period $P_4 = 8$) over 32 consecutive values of n'. Red bars: SubcaseB fires ($F = 0, G = 1$). Blue bars: $F = 0, G = 0$. Gray bars: $F = 1$. Exactly one SubcaseB event per period; the pattern repeats perfectly.

The intuition is left-permutivity: position $k+1$ in c_{n+1} is the *left neighbor* used to compute position k after one CA step. By Equation (2), flipping $c_{n+1}[k+1]$ always flips position k in the next generation. The essential question is whether neighboring cells can *absorb* or *block* this change so it does not reach the center. Crucially, this is *not* automatic from left-permutivity: Rule 90 ($l \oplus r$) is left-permutive but violates lifting (Section 4). The nonlinear term in Rule 30's g-function is essential.

Computationally, no blocking ever occurs for Rule 30: the lifting lemma is verified for all k and all $n \leq 20$ by checking $O(n)$ spike-form initial configurations (not all 2^{2n+1} inputs; see Axiom Inventory in Sec. 3.2), and for $n \leq 3086$ by formal Lean 4 proof using `native_decide` on bounded ranges together with causal-cone structure lemmas (see `LiftingLemma_LeftPermutive.lean`). The remaining open sub-case is the "SubcaseB evenness" condition at $n \geq 3087$, which requires a structural period argument based on the empirically confirmed period structure of Rule 30 diagonals [5]. We further examined all 16 left-permutive elementary CA rules (those of the form $l \oplus g(c, r)$) and find computationally (for $n \leq 20$) that the lifting property co-occurs with all-cells-essential in all 16 cases. The forward direction (lifting $\Rightarrow$ all-cells-essential) is immediate by contrapositive: if some position were not essential, the lifting chain from $n = 1$ would fail at that step. The backward direction (all-cells-essential $\Rightarrow$ lifting) is a computational observation for $n \leq 20$; it is not proved in general for rules other than Rule 30. For Rule 30, both properties hold.

The lifting lemma is taken as an axiom (`lifting_lemma`) in the Lean formalization; see Section 4 for what would be needed to close it.

2.4 Inductive Step

Assume all $2n + 1$ positions are essential at generation n. At generation $n + 1$ there are $2n + 3$ positions:

- Position 0: `left_boundary_essential`.
- Positions $1, \ldots, 2n+1$: for each such position j, position $j-1$ is essential at generation n (inductive hypothesis), and the Lifting Lemma maps it to position j at generation $n+1$.
- Position $2n+2$: `right_boundary_essential`.

Hence all $2n+3$ positions are essential at generation $n+1$.

3 Lean 4 Formalization

The proof is implemented in Lean 4 with Mathlib. The main theorem is:

```
theorem rule30_prize3 (n : Nat) (k : Fin (2 * n + 1)) : Essential n k
```

Table 1 lists the key components.

Lean identifier	Status	Description
`rule30Local_flip_left`	Proved	Left-permutivity
`caStep_flip_blocked`	Proved	Absorption blocking
`backwardFillList_caStep_inverse`	Proved	Backward-fill inverse
`left_boundary_essential`	**Proved**	Left boundary, all n
`right_boundary_essential`	**Proved**	Right boundary, all n
`ts2_last_always_false`	**Proved**	$m = 2$ inactivity, all $n \geq 2$
`rule30n_spikeAt_period`	**Proved**	Parametric F-period lemma
`rule30n_twoSpikeLast_period`	**Proved**	Parametric G-period lemma
`caEvolve_cert_m{4..30}_p{P}`	**Proved**	F-certs, all active $m \leq 30$
`caEvolve_tsl_cert_m{4..22}_p{P}`	**Proved**	G-certs, active $m \leq 22$
`all_cells_essential_by_induction`	Cond.†	Full induction
`rule30_prize3`	Cond.†	$\forall\, n\, k,$ Essential(n, k)
`lifting_lemma`	Open*	Essential$(n, k) \Rightarrow$ Essential$(n{+}1, k{+}1)$

† Proved from boundary theorems + `lifting_lemma` (Path A); no `sorry`. Path B uses `subcaseB_resolution` with a `sorry` for $n' \geq 3087$.

* Axiom in Lean; verified $n \leq 20$ computationally, $n \leq 3086$ by `native_decide`; SubcaseB at $n \geq 3087$ open.

Table 1: Lean formalization status. One open mathematical gap (SubcaseB, $n \geq 3087$); two proof paths each covering it with one axiom.

3.1 Core Definitions

```
def rule30Local (l c r : Bool) : Bool := xor l (c || r)

abbrev Config (n : Nat) := Fin (2 * n + 1) -> Bool

def Essential (n : Nat) (k : Fin (2 * n + 1)) : Prop :=
  exists c : Config n, rule30n n c != rule30n n (flipCell c k)
```

3.2 Axiom Inventory

The formalization contains **two domain axioms** across two proof paths:

- **Path A** (`rule30_prize3` via induction): Uses `lifting_lemma`: Essential$(n, k) \Rightarrow$ Essential$(n{+}1,\ k{+}1)$. Declared as a blanket `axiom` in Lean for all n. The claim has been verified:
  - *Spike-form check* ($n \leq 20$): the lifting property was checked for all k by running the CA on $O(n)$ spike-form configurations (not all 2^{2n+1} inputs); no witness failure was found.
  - *Lean `native_decide`* ($n \leq 3086$): for each n, `native_decide` checks the statement $\forall m \in \mathrm{Fin}(2n{+}3), \ldots$ over $O(n)$ spike configurations, running the CA once per value of m. This is computationally feasible (not exponential). SubcaseA and boundary sub-cases close fully; SubcaseB (even interior positions, $n \geq 3087$) requires a period-structure argument based on Rowland's diagonal-periodicity results [5].
- **Path B** (`rule30_prize3_direct` without lifting): Uses `subcaseB_resolution`: directly asserts the SubcaseB witness exists for all $n' \geq 5$. This axiom covers the same open case as Path A's sorry; it can be eliminated by formalizing the right-boundary independence lemma (the period argument above). `rule30_prize3_direct` also has a `sorry` for $n' \geq 3087$ in the SubcaseB branch.

The `#print axioms rule30_prize3` output (Path A) lists only `lifting_lemma` plus standard Lean/Mathlib axioms; Path B (`rule30_prize3_direct`) lists `subcaseB_resolution` instead.

Both paths have precisely one open mathematical gap: the SubcaseB structure at $n \geq 3087$. The lifting property holds precisely for the 6 left-permutive rules $\{30, 45, 75, 120, 135, 225\}$ (verified $n \leq 20$).

The boundary lemmas (`left_boundary_essential`, `right_boundary_essential`) are *proved theorems*, not axioms. The base case ($n = 0$) is closed by `base_case_n0`, which is trivially true (any Bool value witnesses essentiality at generation 0). Standard Lean/Mathlib axioms (`propext`, `Classical.choice`, `Quot.sound`) are also present.

`lake build P2p.LiftingLemma_LeftPermutive` completes with zero errors (note: `sorry` type-checks in Lean 4, so a clean build does not imply no open sub-cases). Running `#print axioms rule30_prize3` produces:

```
-- axioms used by rule30_prize3:
-- lifting_lemma          (domain conjecture, see Sec. 3.3)
-- propext                (Lean/Mathlib standard)
-- Classical.choice       (Lean/Mathlib standard)
-- Quot.sound             (Lean/Mathlib standard)
```

The only non-standard axiom is `lifting_lemma`. All other dependencies are standard Lean 4/Mathlib axioms present in any Mathlib-using project.

4 Discussion

What is proved. The boundary essentiality results hold for all n with explicit witnesses (`all-zeros` and `only-last-true` configurations) and are machine-verified in Lean 4 with no axioms. Conditional on `lifting_lemma`, the full theorem `rule30_prize3` follows by a clean induction: the inductive step uses only the lifting axiom and the two proved boundary theorems. The *current Lean proof* requires no backward-construction arguments, surjectivity claims, or neighbor-zero invariants; these are discharged by `lifting_lemma`.

What is not proved. The one open item is `lifting_lemma` itself. Left-permutivity (Equation (2)) guarantees that flipping $c_{n+1}[k+1]$ always flips position k in the next generation; the open question is whether there always exists a witness configuration for which this flip is not subsequently absorbed before reaching the center. Computationally, no such absorption occurs for Rule 30 at any generation tested.

The algebraic obstacle is what we term *structured preimage threading*: given a witness c^* at generation n, one must construct a generation-$(n+1)$ preimage c^{**} such that (a) c^{**} extends c^* on the shared cone, and (b) flipping $c^{**}[k+1]$ propagates to the center. The nonlinear $c \vee r$ term makes this threading non-trivial: setting blocker cells to zero simultaneously destroys the witness structure. No such obstruction actually occurs for Rule 30 (verified computationally), but proving it requires controlling the interaction between extension degrees of freedom and the blocker constraint system.

To calibrate: left-permutivity alone does *not* imply lifting. Rule 90 ($l \oplus r$) is also left-permutive, yet lifting *fails*: Essential$(1, 0)$ holds (flipping any left-neighbor always flips the one-step output), but Essential$(2, 1)$ is false, since the 2-step Rule 90 output $c[0]\oplus c[2]\oplus c[4]$ does not depend on $c[1]$ at all. Rule 90 is linear and its evolution skips every other position. Lifting holds (verified computationally for $n \leq 20$) for Rule 30 and the rules in $\{45, 75, 120, 135, 225\}$, and fails for the remaining ten left-permutive rules. The exact algebraic condition separating the two groups is not known in closed form; what is clear is that linearity (as in Rule 90) is sufficient for failure. The failure mode differs structurally: most of the ten rules fail because a specific cell position is *never* essential at any generation (e.g., Rule 90 at odd positions), while Rules 180 and 210 fail specifically at the *center* position at $n = 2$: their two-step composition produces an algebraic cancellation whereby Essential$(1, 1)$ holds but Essential$(2, 2)$ is false, verified exhaustively over all inputs. Computational investigation (2026-03-24, `patterns_ramanujan.md`) now provides an explicit algebraic characterization. For a left-permutive rule $l \oplus f(c, r)$, write f as a Boolean function on $\{0, 1\}^2$. The 16 such rules split into 8 *affine* (where $f(c, r) = ac \oplus br \oplus k$ over GF(2)) and 8 *non-affine*. Among the non-affine rules, two have $f(0, 0) = f(1, 1) = 0$ (i.e., $f = (c \wedge \neg r)$ and $f = (\neg c \wedge r)$); these are the two non-affine rules for which lifting *fails*. All six non-affine rules with $f(0, 0) + f(1, 1) \geq 1$ are exactly $\{30, 45, 75, 120, 135, 225\}$, the rules for which lifting holds.

Diagonal criterion (Conjecture C4, verified $n \leq 6$). The lifting lemma holds for $l \oplus f(c, r)$ if and only if f is non-affine and $f(0, 0) + f(1, 1) \geq 1$.

For Rule 30: $f = c \vee r$, which is non-affine (OR is not XOR-affine), and $f(0, 0) = 0$, $f(1, 1) = 1$, so the diagonal sum is 1. The proof sketch (Conjecture C5) uses the diagonal value $f(1, 1) = 1$: at any generation n, choose a configuration where both boundary neighbors equal 1; then $f(1, 1) = 1$ means the perturbation at position k propagates through the OR gate to position $k + 1$ at generation $n + 1$. This is a finite one-step argument, not dependent on global dynamics. Formalizing this into a complete inductive proof remains open.

The most natural path to closing SubcaseB at $n \geq 3087$ is to exploit the periodicity of the spike-output sequence $F(n, j) := \mathtt{rule30}_n(\mathtt{spike}_j)$. The SubcaseB cases (pairs (n', m) where $\mathtt{spike}_m$ is false but a two-spike witness is true) do *not* cease after $n' = 3086$: for $m = 4$, hard cases appear at $n' = 3093, 3101, 3109, \ldots$ (every 8 steps); for $m = 6$, $m = 14$, and $m = 22$, similarly periodic patterns persist indefinitely. *Re-verification (2026-03-24)*: all SubcaseB positions for $m \in \{4, 6, 8, \ldots, 22\}$ were re-verified using the correct post-loop-50 diagonal-read simulation. Results match prior claims exactly: $m = 4$ at offsets $\{6\}$ (per period 8); $m = 16$ at offsets $\{120, 124, 192\}$ (per period 256); all other active small-m confirmed (136 s total, exhaustive over one full period each).

Right-edge witness approach for $m = 4$ (2026-04-02). The infinite self-similar hierarchy for $m = 4$ Level 3+ (SubcaseB at $n' \equiv 5 \pmod{16384}$) is resolved by a *co-moving frame*: instead of witnesses at fixed positions (which grow without bound as n' increases), use witnesses at fixed offsets from the right edge of the causal cone: $w = 2(n' + 1) - k$ for $k = 10$. As n' grows, w grows with it; the relative structure is constant.

Computationally verified (Python, 2026-04-02): SubcaseB for $m = 4$ fires exclusively at $n' \equiv 5 \pmod 8$ for all $n' \in [3087, 3200]$ (residues $\{0, 1, 2, 3, 4, 6, 7\}$ produce no SubcaseB event). The right-edge function $F_{10}(T) := \texttt{caEvolve}(T, \texttt{spike}(2T{-}10))[\text{center}]$ has period 4 in T; the two-spike function $G_{10}(T) := \texttt{caEvolve}(T, \texttt{twoSpike}(4, 2T{-}10))[\text{center}]$ has period 8. The sensitivity $F_{10}(T) \neq G_{10}(T)$ (i.e., the right-edge witness is valid) has period 8, covering the $n' \equiv 5 \pmod 8$ SubcaseB class.

This collapses the infinite hierarchy to: two focused period-8 axioms (`rightEdgeF_period8_k10`, `rightEdgeG_period8_k10`) stating $F_{10}(T{+}8) = F_{10}(T)$ and $G_{10}(T{+}8) = G_{10}(T)$ for $T \geq 100$, together with one native_decide base case ($T = 3094$, corresponding to $n' = 3093 \equiv 5 \pmod 8$). The formalization is in `P2p/SubcaseB_m4_RightEdge.lean` (0 sorries, 2 axioms). The two period-8 axioms are closeable via 41-cell, 8-step combined-pattern period certificates (feasible `native_decide`).

Period structure and max-m growth. The existing Lean lemmas in `CausalConeLemmas.lean` establish that the *open-boundary* (causal-cone) evolution starting from a spike at position 6 has period 16, and starting from position 20 has period 256; these are certified by `native_decide` on finite state spaces. Empirical computation confirms that the SubcaseB condition for all eight active positions $m \leq 20$ is periodic in n' with period at most 256.

However, the spike at position $m = 36$ is first verified to satisfy the SubcaseB condition at $n' = 4113$ (offset $1026 = 4 \times 256 + 2$ from base 3087), which is *not* in the first 256-step window. The full period has been determined by direct computation: hits at $n' \in \{4113, 4117, 8209\}$ are followed by $\{20497, 20501, 24593\} = \{4113, 4117, 8209\} + 16384$, confirming period exactly $\mathbf{16384 = 2^{14}}$ (not the earlier estimate of 4096). More precisely, the gap from the first cluster $\{4113, 4117\}$ to the singleton $\{8209\}$ is $4096 = 2^{12}$, but the *cluster repeat period* is $16384 = 2^{14}$. This fits the doubling law: $m = 34$ has period $8192 = 2^{13}$ and $m = 36$ has period 2^{14}. The doubling continues: $m = 38$ has period $32768 = 2^{15}$ (first SubcaseB hits at $n' = 8210, 8214$; confirmed by 132-point period check), with the active sequence $\{\ldots, 30, 34, 36, 38\}$ yielding periods $\{4096, 8192, 16384, 32768\}$. For $m = 34$ the SubcaseB hit set within one period is *exactly* $\{4112, 4116\}$: an exhaustive G-check of all 4096 $F{=}0$ candidates in $[3087, 11279)$ was run (loop-57, 2026-03-24; 659 s), confirming no additional hits. The second period gives $\{12304, 12308\} = \{4112, 4116\} + 8192$, spot-checked directly. For $m = 36$, an analogous exhaustive G-check of all 946 $F{=}0$ candidates in $[3087, 5000)$ was run (loop-58, 2026-03-24; 53 s): SubcaseB occurs at exactly $\{4113, 4117\}$ and nowhere else in this window. The singleton $n' = 8209$ and all three period-repeat hits $\{20497, 20501, 24593\} = \{4113, 4117, 8209\} + 16384$ were confirmed by direct computation. *Independent adversarial re-verification (2026-03-*

25): $m = 34$ first hit at $n' = 4112$ confirmed $(F, G) = (0, 1)$; $m = 36$ at $n' = 4113$ confirmed; $m = 38$ at $n' = 8210$ confirmed; second-period hits $\{12304, 12308\}$ for $m = 34$ and $\{20497, 20501, 24593\}$ for $m = 36$ all confirmed $(F, G) = (0, 1)$ by direct simulation. The period-256 claim therefore does not extend to the full SubcaseB set. Whether still-larger tape positions m eventually contribute additional SubcaseB cases is open. Comprehensive evidence for inactivity above $m = 38$ (loop-28, 2026-03-24):

- Triangle-method scans (loop-24/28) confirm *0 SubcaseB* for all even $m \in [44, 80]$ over $n' \in [3087, 45001)$ (12 s, 20 000+ F=0 candidates per m, first-10 G-checked). Upgraded: loop-63 (2026-03-24) exhaustively G-checks *all* F=0 candidates in $[3087, 3500)$ for $m = 40, 42, \ldots, 80$ (21 values), 0 SubcaseB.

- $m = 40$: F-period certified = 65536 (loop-27); first 100 of 32828 F=0 candidates in one period individually G-checked: 0 SubcaseB.

- $m = 42, \ldots, 200$: triangle method (loop-24) to $n' = 20001$.

Caution: an earlier "step-100 sweep $[3087, 45000)$" cited here is not valid evidence: step-100 detects *0 of 4* SubcaseB events for active $m = 34$, *0 of 6* for $m = 36$, and *0 of 4* for $m = 38$ in the same range (loop-28 analytic verification: all known SubcaseB events have $n' \not\equiv 87 \pmod{100}$, the step-100 grid offset). Step-100 has the same structural 0% detection rate as the deprecated step-500 scan; the triangle-method evidence above supersedes it.

Active and inactive positions. Not every even tape position can contribute to SubcaseB. Define $F(n, m)$ and $G(n, m)$ as above. Computational analysis of $(F(n, m), G(n, m))$ reveals a sharp dichotomy:

- *Active*: $M_{\text{act}} = \{4, 6, 8, 10, 12, 14, 16, 20, 22, 24, 26, 28, 30, 34, 36, 38\}$. The $(0, 1)$ pair occurs; periods range from 8 to 32768 (P_m not strictly doubling: $P_{12} = P_{14} = 64$ and $P_{20} = P_{22} = 256$ are *pre-plateau flat*, i.e. $d_{\text{actual}} \leq 4$); period minimality confirmed for all $m \leq 38$: $m \leq 28$ by full F-sequence comparison (loop-20); $m \in \{30, 34, 36, 38\}$ by triangle method showing $P/2$ fails at $n' = 3087$ (loop-25, 2026-03-24)). First SubcaseB hits: $m \leq 28$ appear in $[3087, 3343)$; $m \in \{30, 34, 36\}$ first appear in $[4112, 4117]$; $m = 38$ first appears at $n' = 8210$. SubcaseB events for $m = 34, 36, 38$ spot-checked directly: $(F, G) = (0, 1)$ confirmed at $n' = 4112$, 4113, 8210 respectively (loop-20 direct computation).

- *Inactive* ($(0, 1)$ never occurs): $m \in \{2, 18, 32\} \cup \{40, 42, \ldots\}$. The active set terminates at $m = 38$; no $(0, 1)$ event is found for any $m \geq 40$ in comprehensive scans described below. Completeness of M_{act} within $[4, 38]$ is verified: $m = 18$ and $m = 32$ are confirmed inactive, all other even $m \in [4, 38]$ are confirmed active, all by direct SubcaseB scan (loop-20). *Independent adversarial re-verification (2026-03-25)*: triangle-method F-scan for $n' \in [3087, 3600)$ followed by exhaustive G-check of all F=0 candidates confirms: $m = 18$ has 256 F=0 candidates and *zero* SubcaseB; $m = 32$ has 250 F=0 candidates and *zero* SubcaseB; $m = 20$ has first SubcaseB at $n' = 3341$ (offset 254, exactly matching period-256 structure); $m = 22$ first SubcaseB at $n' = 3342$ (offset 255); $m = 40$ has 251 F=0 and zero SubcaseB. All five critical paper claims confirmed by independent computation.

For $m = 2$, inactivity is formally proved in Lean (`ts2_last_always_false`). The inactive positions exhibit two distinct behaviors. *Weakly inactive* ($F{=}G$ fails occasionally, but $(0,1)$ is absent): $m = 18$ has $(1,0)$ events at $n' \in \{3280, 3336, 3340, \ldots\}$; $m = 32$ has $(1,0)$ at $n' \in \{3347, 4111, 4115\}$ within a comprehensive $[3087, 6000)$ scan (2913 values, zero $(0,1)$)—distinguishing $m = 32$ from $m = 36$ (first active hit at $n' = 4113$; second at $n' = 4117$; both within the same window; confirmed by adversarial loop 1774432000); $m = 38$ is active: first $(0,1)$ at $n' = 8210$ (period 32768, verified). *Weakly inactive* ($I = 0$ only when $F = 1$): $m = 40$ has a single $(1,0)$ at $n' = 36887$ and zero $(0,1)$ in $[3087, 110000)$, including at all resonant positions $\{8211, 16403, 32787\}$ where last$-m \in \{2^{14}, 2^{15}, 2^{16}\}$ (giving $(1,1)$, $(0,0)$, $(1,1)$ respectively; none is SubcaseB). $P_{40} = 65536$ confirmed (adversarial 2026-03-26: period-65536 holds for first 500 F-values, period-32768 fails). *Caution (adversarial 2026-03-26)*: the existing evidence for $m = 40$ inactivity covers only the first 100 of ≈32,828 $F{=}0$ candidates in one period (0.3% coverage), plus the resonance window $[13233, 13833)$ (294 candidates, G-checked exhaustively, zero SubcaseB; this targets the predicted first-hit offset by analogy with $m = 38$'s first SubcaseB at offset 5123). A rigorous inactivity proof for $m = 40$ requires exhaustive G-check of all ≈32,828 $F{=}0$ candidates in one full period $[3087, 3087{+}65536)$; this check is feasible (≈4 h) but has not yet been run. *Strictly $F{=}G$ for $n' \geq 3087$* (only $(0,0)$ and $(1,1)$ observed in the large-n' regime): $m = 42, \ldots, 200$ confirmed by triangle-method dense scans (loop-24, 2026-03-24): $m = 42, \ldots, 80$ over $[3087, 20001)$ with exhaustive G-check on all $F{=}0$ candidates in $[3087, 7000)$; $m = 82, \ldots, 200$ over $[3087, 20001)$. $P_{42} = 131072$ confirmed (adversarial 2026-03-26: 200-point spot-check at offsets 0 and 131072 match; half-period 65536 fails). Resonance window for $m = 42$: predicted first hit at $n' {\approx} 23579$; exhaustive G-check of all 293 $F{=}0$ candidates in $[23479, 24079)$ yields zero SubcaseB (adversarial 2026-03-26). *Caution (adversarial loop 1774416XXX, 2026-03-25)*: for $n' < 3087$, inactive $m \equiv 4$ or $6 \pmod 8$ (e.g. $m = 44, 46, 52, 54, \ldots$) do exhibit $(0,1)$ at the isolated point $n' = m/2 + 3$; these small-n' events are covered by Lean's `native_decide` and do not affect the prize proof. SubcaseB is structurally impossible for inactive positions *at large n'*, and $m = 32$ in particular is very strongly consistent with permanent inactivity. A formal proof requires establishing $(0,1)$ is excluded for *all* $n' \geq 3087$, which is itself a periodicity argument.

Proof plan. A formal SubcaseB closure has three parts.

Part A – inactive positions. For $m = 2$: already proved. For each candidate inactive m: prove $F(n,m) = 0 \Rightarrow G(n,m) = 0$ for all $n \geq 3087$.

A structural lemma clarifies the algebraic relationship. Define $H(n')$ as the center value after $n'{+}1$ steps from a spike at `last` $= 2n'{+}2$ alone.

Lemma 4.1 (Last-spike lemma)**.** *$H(n') = 1$ for all $n' \geq 0$.*

Proof sketch. Frontier invariant: at step j, all cells at positions $< 2n'{+}2 - j$ are 0 (proved by induction: the rule 30 update of an all-zero left/center/right triple is 0). The frontier cell at step j (position $2n'{+}2 - j$) has left and center neighbors 0 at step $j{-}1$, so its value equals its right neighbor at step $j{-}1$ (position $2n'{+}3 - j$), which is the previous frontier. By the chain $H(n') = \text{cell}[n'{+}2]_{@n'} = \text{cell}[n'{+}3]_{@n'-1} = \cdots = \text{cell}[2n'{+}2]_{@0} = 1$. □

Since $G = F \oplus H \oplus I$ (where I is the nonlinear interaction term) and $H = 1$, we have $G = F \oplus 1 \oplus I$. SubcaseB ($F = 0, G = 1$) $\Leftrightarrow I = 0$. Defining $I(n', m) := F \oplus G \oplus 1$, the proof targets become:

- *Weakly inactive*: for $m = 40$ (single $(1, 0)$ at $n' = 36887$, zero $(0, 1)$ in $[3087, 110000)$; resonance test at $n' = 16403$ gives $(0, 0)$, confirming inactivity). Same proof target as $m = 18, 32$: show $F = 0 \Rightarrow I = 1$.

- *Strictly F=G*: for $m = 42, \ldots, 200$ (even), $I(n', m) = 1$ for all checked n'—equivalently $G = F$. Verified by the causal-cone triangle method (2026-03-24): $m = 42, \ldots, 80$ confirmed over $[3087, 20001)$ with exhaustive G-check on all $\approx 35\,000$ $F{=}0$ candidates in $[3087, 7000)$; $m = 82, \ldots, 200$ confirmed over $[3087, 20001)$ with first-100 G-check per m (loop-34, 2026-03-24; upgraded from loop-24's first-20); upgraded to exhaustive G-check on all $F{=}0$ candidates in $[3087, 3500)$ (loop-67, 2026-03-24, C tool, 12 s, zero SubcaseB); adversarial extension (2026-03-26): $F{=}G$ verified for $m = 82, 84, 86$ over the extended range $[3087, 3600)$ (direct Python computation, zero violations). *Adversarial caveat*: for $n' > 3600$ and $m \geq 82$, only first-100 G-check sampling applies; a fully rigorous inactivity proof requires either the Φ_3-fingerprint structural argument or exhaustive period-based reduction. The proof target is $I \equiv 1$, i.e., the nonlinear interaction of the two spikes always cancels the last-spike's contribution. (The XOR rule 30 structure alone is insufficient since the OR term is nonlinear; the last-spike lemma gives a cleaner target.)

- *Weakly inactive* (I occasionally $= 0$, but $F = 1$ whenever $I = 0$): for $m = 18$ and $m = 32$ only. The (F, G) patterns are periodic: $m = 18$ has period 256 (exact; verified by checking $(F(n'{+}256), G(n'{+}256)) = (F(n'), G(n'))$ for all $n' \in [3087, 3343)$ and period 128 fails; period minimality confirmed by adversarial loop-20 full-sequence scan), with $(1, 0)$ triplets at offsets $\{193, 249, 253\}$ per period and *zero* $(0, 1)$ verified directly in two full periods $[3087, 3599)$ (the remaining 25 of 27 claimed periods follow from the exact period-256 structure). $m = 32$ has period 4096 (same as active $m = 30$; confirmed by cluster $\{3347, 4111, 4115\}$ repeating at $+k{\cdot}4096$ for $k = 1, 2, 4, 8$), with *zero* $(0, 1)$ verified directly in the full first period $[3087, 7183)$ (loop-20 direct scan, 197s; adversarial re-verification 2026-03-26: all 2034 $F{=}0$ candidates in one period exhaustively G-checked, 245 s, zero SubcaseB, rigorously confirming permanent inactivity via periodicity). The cluster $\{3347, 4111, 4115\}$ consists of $(F, G) = (1, 0)$ events (not SubcaseB): independently verified $F = 1$, $G = 0$ at each, and $F = 1$ at $+4096$ and $+8192$ repeats (adversarial 2026-03-26). $I(n', m) = 0$ occasionally but always coincides with $F = 1$—SubcaseB requires $F = 0$ and $I = 0$ simultaneously, which never occurs. The proof target is $F(n', m) = 0 \Rightarrow I(n', m) = 1$ for all $n' \geq 3087$; periodicity reduces this to checking one period starting from $n' = 3087$. *Inactivity for $m = 32$ is thus rigorously established computationally* (exhaustive scan of one full period; formal proof by period-certificate reduction remains open but the computational gap is closed).

Part B – fixed-m active positions. For M_{act} (the 16 active positions), the cluster period P_m grows with m (empirical):

$$m = 4 \to 8,\ m = 6 \to 16,\ m = 8 \to 32,\ m \in \{10, 12, 14\} \to 64,\ m \in \{16, 20, 22\} \to 256,$$

$$m = 24 \to 512,\ m = 26 \to 1024,\ m = 28 \to 2048,\ m = 30 \to 4096,$$

$$m = 34 \to 8192,\ m = 36 \to 16384,\ m = 38 \to 32768.$$

All periods are verified by direct computation and period-minimality is confirmed: period $P_m/2$ fails for every active m with $4 \le m \le 38$. For $m \le 28$, full F-sequence comparison over a complete period was run (loop-20). A caution: for $m = 26$ and $m = 28$, the value $F(3087, m)$ happens to equal $F(3087{+}P/2, m)$ by coincidence, so single-point spot checks are insufficient; full-sequence scans were needed. For $m \in \{30, 34, 36, 38\}$, the triangle method (2026-03-24) confirms that $P/2$ fails at the very first check point $n' = 3087$:

- $m = 30$: half-period $P/2 = 2048$ fails at $n' = 3087$; $F(3087, 30) = 1 \ne 0 = F(5135, 30)$.
- $m = 34$: half-period $P/2 = 4096$ fails at $n' = 3087$; $F(3087, 34) = 1 \ne 0 = F(7183, 34)$.
- $m = 36$: half-period $P/2 = 8192$ fails at $n' = 3087$; $F(3087, 36) = 0 \ne 1 = F(11279, 36)$.
- $m = 38$: half-period $P/2 = 16384$ fails at $n' = 3087$; $F(3087, 38) = 1 \ne 0 = F(19471, 38)$.

Each full-period spot-check (4096 points) passes, confirming P is correct. Moreover, for $m \in \{34, 36, 38\}$ the F-period certificates confirm minimality (loop-39, 2026-03-24): the period-P_m certificate holds while the period-$P_m/2$ certificate fails at position 0 in all three cases (≤ 0.3 s each via numpy).

The period structure has two regimes. For $m \ge 24$: a *clean doubling* holds with each successive active position, irrespective of inactive positions. Active $m \in \{24, \dots, 38\}$ skip the inactive $m = 32$ and give $P_m = 2^9, 2^{10}, \dots, 2^{15}$ in order, verified by direct computation. $m = 36$: SubcaseB hits confirm period 2^{14}; $m = 38$: period 2^{15} confirmed by 132-point check (loop-30 exhaustively G-checked all 2947 $F{=}0$ candidates in $[3087, 9000)$; loop-61, 2026-03-24, re-verified SubcaseB at $n' = 8210, 8214$ with the correct simulation and confirmed the period-32768 repeat at $n' = 40978, 40982 = \{8210, 8214\} + 32768$).

For $m < 24$: the pattern is irregular with two *plateaus*. $m \in \{10, 12, 14\}$ all share period $64 = 2^6$; $m \in \{16, 20, 22\}$ all share period $256 = 2^8$ (a factor-of-4 jump from 64; inactive $m = 18$ also has period 256, verified directly over two full periods $[3087, 3599)$ and confirmed minimal by period-128 failure; the remaining periods to $n' = 10000$ follow from the exact period structure)—so the entire block $m \in \{16, 18, 20, 22\}$ operates at period 256, with active positions producing $(0, 1)$ and $m = 18$ producing only $(1, 0)$). Similarly, inactive $m = 32$ has period 4096, matching active $m = 30$, with zero $(0, 1)$ verified directly in the full first period $[3087, 7183)$. The $m = 16$ cluster has a more complex internal structure: three hits per period at offsets $120, 124, 192$ (measured from $n' = 3087$) with gaps $(4, 68, 184)$ summing to 256 (re-verified with correct post-loop-50 simulation, 2026-03-24; all offsets confirmed). The doubling law does *not* hold universally across all active m. A striking corollary: the value $\log_2 P_m = 7$ (period 128) is *absent* from the sequence $\{\log_2 P_m : m \in M_{\text{act}}\} = \{3, 4, 5, 6, 6, 6, 8, 8, 8, 9, \dots, 15\}$. This is the unique gap in $[3, 15]$ and serves as the algebraic fingerprint of $m = 18$'s inactivity: the plateau window $\{16, 18, 20, 22\}$ has period $256 = 2^8$, skipping 2^7. (Verified: universal F-period doubling $P_{m+2} = 2P_m$ holds for all even $m \in [2, 42]$ with exactly two plateau windows at $m \in [10, 14]$ and $m \in [16, 22]$; iteration-4, 2026-03-24. Adversarial re-verification 2026-03-26: $P_{40} = 65536$ and $P_{42} = 131072$ independently confirmed, extending the verified range to $m \in [2, 42]$. The resonance-window SubcaseB check for both $m = 40$ and $m = 42$ (exhaustive G-scan of $\approx$294 candidates each near the predicted first-hit position) yields zero SubcaseB, consistent with inactivity.)

SubcaseB transient structure (adversarial loops 1774413233 & 1774416XXX, 2026-03-25). All even m exhibit SubcaseB for small n'. The transient window has two contributions: (i) a *trivial range* $n' \le m/2-2$ where m lies outside the causal cone (so $F=0$ automatically and $G=H(n')=1$); and (ii) for $m \equiv 4$ or $6 \pmod 8$, one additional *genuine* SubcaseB event at $n'=m/2+3$ (verified for $m \le 100$, adversarial loop 2026-03-25). The earlier formula "$\text{last}_{\text{SB}} = m/2-2$" was incorrect for $m \equiv 4, 6 \pmod 8$: the correct last transient event for those m is $n'=m/2+3$. For inactive m, SubcaseB does *not* recur after these transient events. Active m, by contrast, have SubcaseB recurring periodically for $n' \ge 3087$ (the large-n' regime), regardless of their small-n' transient structure. The transient-vs-recurrent dichotomy is the sharpest characterization of M_{act} found by direct computation.

Binomial-machine structure (Ramanujan analysis, 2026-03-24). The F-sequences, viewed as binary strings of length P_m over GF(2), have a striking algebraic regularity. For each active $m \in M_{\text{act}}$, the Berlekamp-Massey connection polynomial of F_m is exactly $(x+1)^{L_m}$, where L_m is the linear complexity (independently verified for $m \le 30$; BM requires BASE $\gg 2L_m$ samples to converge). Equivalently, the L_m-th forward difference $\Delta^{L_m}[F_m]=0$ and $\Delta^{L_m-1}[F_m]=\mathbf{1}$ (all ones), so L_m is sharp. This means each F-sequence admits a binomial representation $F_m(n') = \sum_{k=0}^{L_m-1} a_k \binom{n'}{k} \pmod 2$, the same algebraic form as Pascal's triangle mod 2 (Rule 90). Thus Rule 30, despite its nonlinear $c \vee r$ term, generates diagonal sequences whose GF(2) algebra is governed by the same annihilator family as the linear Rule 90. The linear complexity follows the pattern: $L_m = P_m/2+1$ at each new period-plateau start (deficit $P_m/2-1$), decaying by 1 per successive active m within a doubling regime. The extended LFSR table (adversarial loop 1774413233, 2026-03-24) gives: $L = 5, 9, 26, 59, 64, 64, 129, 256, 254, 257, 770, 1795, 3844$ for $m = 4, 6, 8, 10, 12, 14, 16, 20, 22, 24, 26, 28, 30$ respectively. *Defect formula (Ramanujan loop 1774417929, 2026-03-25)*: Within the strictly-increasing-period segment $m \in \{24, 26, 28, 30\}$, the defect $d_m = P_m - L_m$ satisfies $d_m = 267 - m/2$ (i.e., $255, 254, 253, 252$), an exact arithmetic progression with step -1. This yields the prediction $L_{34} = 4097$, $L_{36} = 12290$, $L_{38} = 28675$ for Run C $m \in \{34, 36, 38\}$. *Verification (adaptive loop 1774421888, 2026-03-25)*: $L_{34} = 4097 = P_{34}/2+1$ is **confirmed** using $N = 8600$ BM samples at BASE$=3087$. Connection polynomial $(1+x)^{4097} = 1+x+x^{4096}+x^{4097}$, confirming the plateau-start pattern (same as $m = 4, 6, 16, 24$). *Run-C LFSR table completed (adaptive loop 1774422600, 2026-03-25)*: Using $N = 25500$ and $N = 58500$ BM samples respectively, the actual values are:

$$L_{36} = 8193 = P_{36}/2+1, \quad L_{38} = 24578 = P_{38} - 8190.$$

The prediction $L_{36} = 12290$ was **wrong**: $m = 36$ is a *second* plateau-start, not a continuation of the arithmetic progression from $m = 34$. Connection poly $(1+x)^{8193} = 1+x+x^{8192}+x^{8193}$ for $m = 36$; $(1+x)^{24578} = (1+x^2)(1+x^{8192})(1+x^{16384})$ for $m = 38$. Defect $d_{38} = 8190 = d_{36}-1$ confirms the step-(-1) arithmetic progression *restarting* from the $m = 36$ plateau start. Complete Run-C defect table: $d_{34} = 4095$, $d_{36} = 8191$, $d_{38} = 8190$. *Run-D: inactive $m = 40$ (adversarial loop 1774432000, 2026-03-25)*: with $P_{40} = 65536$ confirmed, BM with $N = 140000$ samples ($> 2L$) gives:

$$L_{40} = 57347, \quad d_{40} = 65536 - 57347 = 8189 = d_{38} - 1.$$

The step-(-1) progression continues through inactive $m = 40$, exactly as the period-doubling law does. The connection polynomial $(1+x)^{57347} = (1+x)(1+x^2)(1+x^{8192})(1+x^{16384})(1+x^{32768})$ has $2^5 = 32$ nonzero positions in five groups of 4 (first 8: $\{0, 1, 2, 3, 8192, 8193, 8194, 8195\}$); spacing $= 1$, not decimated. *Run-E: inactive $m = 42$ (adversarial loop 1774435000, 2026-03-25)*: BM with $N = 250000$ samples ($> 2 \times 122884 = 245768$) **confirms** $d_{42} = 8188$, $L_{42} =$

122884, connection poly $(1+x)^{122884} = (1+x^4)(1+x^{8192})(1+x^{16384})(1+x^{32768})(1+x^{65536})$ (first 8 nonzero positions $\{0, 4, 8192, 8196, 16384, 16388, 24576, 24580\}$, spacing$=4$, inner$=30721$, not Mersenne). Step-$(-1)$ check: $d_{42} - d_{38} = -2$ (expected -2); $d_{42} - d_{40} = -1$ (expected -1). ✓ The complete defect table for $m \in \{36, 38, 40, 42\}$ is $\{8191, 8190, 8189, 8188\}$, an exact arithmetic progression with step -1, traversing two plateau starts, one active, and one block-inactive position. *Run-F: inactive $m = 44$ (adversarial loop 1774444000, 2026-03-25)*: BM with $N = 515000$ ($> 2\times 253957 = 507914$) **confirms** $d_{44} = 8187$, $L_{44} = 253957$. Step-(-1) check: $d_{44} - d_{42} = 8187 - 8188 = -1$. ✓ Extended defect table: $m \in \{36, 38, 40, 42, 44\}$ gives $\{8191, 8190, 8189, 8188, 8187\}$.

Synthesis: three LFSR defect regimes (adversarial SYNTHESIZE loop 1774433717-C, 2026-03-25). The defect sequence $d_m = P_m - L_m$ is governed by three distinct regimes, not a single universal law:

1. **Run-B regime** ($m \in \{24, 26, 28, 30\}$): $d_m = 267 - m/2$, giving $\{255, 254, 253, 252\}$. Step-(-1) holds within this block only.

2. **Run-C/D/E regime** ($m \geq 36$, i.e., after the second plateau start): $d_m = 8191 - (m - 36)/2$, giving $\{8191, 8190, 8189, 8188, \ldots\}$ for $m \in \{36, 38, 40, 42, \ldots\}$. Step-$(-1)$ holds here regardless of active/inactive status. This regime *restarts* at the $m = 36$ plateau ($d_{36} = 8191 = P_{36}/2 - 1$); $m = 34$ has $d_{34} = 4095 = P_{34}/2 - 1$ (its own plateau formula, *not* Run-B).

3. **Flat-chain regime** ($m \in \{12, 14, 18, 20, 22, 32\}$): $d_m \in \{0, 1, 2, 4\}$. Step-(-1) does *not* hold; flat-chain m have anomalously small defects unrelated to the doubling-regime formulas.

The plateau starts $m \in \{4, 6, 16, 24, 34, 36\}$ act as regime boundaries, each resetting $d_m = P_m/2 - 1$. Between resets, step-(-1) progresses until either the next plateau start or the flat chain interrupts. For example, starting from the $m = 6$ plateau ($d_6 = 7$): $d_8 = 6$, $d_{10} = 5$ (two steps of step-(-1)), then the flat chain $\{12, 14\}$ interrupts (both have $d = 0$, breaking the progression) before the $m = 16$ plateau resets. This corresponds to the *pre-Run-B* regime of two doubling steps followed by two flat-chain positions. (Note: $m = 4$ and $m = 6$ are both plateau starts, not doubling steps.)

Plateau-start $\log_2 P$ pattern: plateau starts occur at $\log_2 P \in \{3, 4, 8, 9, 13, 14\}$, i.e. consecutive pairs separated by gaps of 4, with alternating differences $(+1, +4)$ in the sequence. *Caution (corrected loop 1774426000, 2026-03-25)*: an earlier conjecture that $\log_2 P \equiv 3$ or $4 \pmod 5$ would predict plateau starts is **false**: $m = 20, 22$ both have $\log_2 P = 8 \equiv 3 \pmod 5$ but are not plateau starts. The $(+1, +4)$ pattern is a descriptive observation about known plateau starts, not a predictive algebraic criterion. *Adversarial confirmation (2026-03-25)*: the pattern predicts the next pair at $\log_2 P \in \{18, 19\}$, i.e. $m = 44$ would be the next plateau start. This is **false**: a complement-half test at 20 sample positions gives only 10/20 complements (not 20/20), definitively ruling out $m = 44$ as a plateau start. A BM probe with $N = 280000$ shows $L \approx N/2$ (not converged); full BM with $N = 515000$ **confirms** $L_{44} = 253957$, $d_{44} = 8187 = d_{42} - 1$ (adversarial loop 1774444000, 2026-03-25, total time 3634 s). Connection polynomial first 12 nonzero positions: $\{0, 1, 4, 5, 8192, 8193, 8196, 8197, 16384, 16385, 16388, 16389\}$, spacing 1 within clusters of 4, inter-cluster spacing $8192 - 5 = 8187$. *Run-F:* The step-(-1) progression $d_m = 8191 - (m - 36)/2$ now extends through $m \in \{36, 38, 40, 42, 44\}$: defects

$\{8191, 8190, 8189, 8188, 8187\}$, all confirmed. The next plateau start, if any, lies strictly beyond $m = 44$.

Unified LFSR formula (Explore loop 1774460000, 2026-03-25). All six known doubling regimes are governed by a single formula:

$$L_m = 2^b\big(2^j - 1\big) + j, \qquad d_m = P_m - L_m = 2^b - j,$$

where b is the *plateau b-value* ($b = \log_2(P_{\text{plateau}}/2)$) and $j = (m - m_0)/2 + 1$ counts steps from the plateau start m_0. Parameter table: $(b, m_0) \in \{(2, 4), (3, 6), (7, 16), (8, 24), (12, 34), (13, 36)\}$. **All 15 known L_m values pass** (verified computationally): $L_4 = 5$, $L_6 = 9$, $L_8 = 26$, $L_{10} = 59$, $L_{16} = 129$, $L_{24} = 257$, $L_{26} = 770$, $L_{28} = 1795$, $L_{30} = 3844$, $L_{34} = 4097$, $L_{36} = 8193$, $L_{38} = 24578$, $L_{40} = 57347$, $L_{42} = 122884$, $L_{44} = 253957$. **Corollaries.** (i) Step-(-1): j increments by 1 per $m+2$, so d decrements by 1. (ii) Plateau reset: new run starts at $j = 1$, giving $d = 2^b - 1 = P/2 - 1$ exactly. (iii) Inter-cluster spacing in the connection polynomial equals $d_m = 2^b - j$ (verified for $m = 44$: spacing $= 8192 - 5 = 8187 = d_{44}$). (iv) Flat-chain m (*e.g.* $m \in \{12, 14, 18, 20, 22, 32\}$) do *not* follow this formula; their anomalously small $d \leq 4$ signals a different algebraic regime. The b-value sequence $\{2, 3, 7, 8, 12, 13\}$ for the six plateau starts satisfies the same $(+1, +4)$ alternating-difference pattern as $\log_2 P$, confirming $b = \log_2(P/2)$ at each plateau start.

Correction (Ramanujan loop 1774417929, 2026-03-25): the earlier claim that inactive m "do not have the $(1 + x)^L$ structure" was based on insufficient sample size ($N = 300 < 2L_m$). With $N = 512 > 2 \cdot 255 = 510$ samples, BM for $m = 18$ converges to $L_{18} = 255 = P_{18} - 1$, confirming that the F-sequence for inactive $m = 18$ is *also* $(1 + x)^{255}$—a binomial sequence. Within the $P = 256$ plateau $\{16, 18, 20, 22\}$, the linear complexities are: $L_{16} = 129$, $L_{18} = 255$, $L_{20} = 256$, $L_{22} = 254$. Inactive $m = 18$ has *higher* linear complexity than active $m = 22$! *All* even m have F-sequences with $(1 + x)^L$ minimal polynomial; the active/inactive distinction lies not in L but in whether the interaction term $I(n', m) := F \oplus G \oplus 1$ ever equals 0 at large n' (active: yes, periodically; inactive: $I \equiv 1$ for all $n' \geq 3087$). A valid GF(2) algebraic fingerprint exists at the *indicator polynomial* level: the generating polynomial $\sum_{m \in \text{inactive} \cap [0,38]} y^{m/2} = 1 + y + y^9 + y^{16}$ is divisible by $\Phi_3(y) = 1 + y + y^2$ over GF(2), while the active-set polynomial is not (Ramanujan loop 1774417929, 2026-03-25).

Fermat denominator theorem (2026-03-24). For each active m at the *start of a new period level* ($L_m = P_m/2 + 1$), define $\alpha_m = \sum_{k=0}^{P_m - 1} F(n_0 + k, m) \cdot 2^k$ (any starting n_0). The fraction $\alpha_m/(2^{P_m} - 1)$ in lowest terms has denominator *exactly* $2^{P_m/2} + 1$ (a Fermat number): $m \in \{4, 6, 16, 24\}$ give denominators 17, 257, $2^{128} + 1$, $2^{256} + 1$. The *mechanism* (Ramanujan analysis, 2026-03-24; verified loop 1774426000, 2026-03-25): period-start F_m-sequences satisfy the *complement-half condition* $F(n' + P_m/2, m) = 1 - F(n', m)$ for all n', i.e., the first and second half-periods are bitwise complements. This condition is an *exact characterization*: among all active $m \in M_{\text{act}}$ with $m \leq 38$, the condition holds *if and only if* m is a plateau start ($m \in \{4, 6, 16, 24, 34, 36\}$). All nine non-plateau active m fail the condition (e.g. $m = 8$: fraction of anti-symmetric positions at lag $P/2 = 16$ is 0.625, not 1.0). Writing $\beta = \sum_{k=0}^{P_m/2 - 1} F(n_0 + k, m) \cdot 2^k$, one computes $\alpha_m = (2^{P_m/2} - 1)(2^{P_m/2} - \beta)$, hence

$$\frac{\alpha_m}{2^{P_m} - 1} = \frac{(2^{P_m/2} - 1)(2^{P_m/2} - \beta)}{(2^{P_m/2} - 1)(2^{P_m/2} + 1)} = \frac{2^{P_m/2} - \beta}{2^{P_m/2} + 1},$$

giving denominator $2^{P_m/2} + 1$ when $\gcd(2^{P_m/2} - \beta,\, 2^{P_m/2} + 1) = 1$. Non-starts lack this complement-half symmetry and accumulate multiple cyclotomic factors (e.g., $m = 8$: den $= 17 \times 257 \times 65537 = F_2 \cdot F_3 \cdot F_4$).

F-period certificates (Python-verified, 2026-03-24). The Lean-style certificate

$$\texttt{caEvolve}\ P_m\ (\texttt{spike}_m(2P_m{+}2m{+}1))\ =\ \texttt{spike}_m(2m{+}1)$$

holds for $m \in \{18, 32\}$ (loop-27: $P = 256$ and $P = 4096$ respectively, both minimal). The sporadic inactive positions $m \in \{18, 32\}$ are *not* the only even m for which the period-doubling law *fails*. *Correction (adversarial loop 1774435000, 2026-03-25)*: the claim "all other even m obey $P_{m+2} = 2P_m$" is incorrect. A rigorous 3000-pair period test reveals two *pre-plateau flat* regions where the period stops doubling:

- $m \in \{12, 14\}$: $P_{12} = P_{14} = 64 = P_{10}$ (flat at 64; period-doubling would give $P_{12} = 128$, $P_{14} = 256$). Both $m = 12, 14$ are *active.*
- $m \in \{20, 22\}$: $P_{20} = P_{22} = 256 = P_{18} = P_{16}$ (flat at 256; period-doubling would give $P_{20} = P_{22} = 512$). Both $m = 20, 22$ are *active.*

Together with the known flat sporadic positions $P_{18} = P_{16} = 256$ (inactive) and $P_{32} = P_{30} = 4096$ (inactive), the complete set of period-doubling exceptions is $m \in \{12, 14, 18, 20, 22, 32\}$. These form two "pre-plateau flat" chains ending just before each plateau start at $m = 16$ ($P = 256$) and $m = 24$ ($P = 512$): plateau $m = 16$ is preceded by the flat chain $\{12, 14\}$; plateau $m = 24$ is preceded by the flat chain $\{18, 20, 22\}$; plateau $m = 34$ is preceded by the single flat $\{32\}$. The prize proof is unaffected: the active-m periods are $\{8, 16, 32, 64, \mathbf{64}, \mathbf{64}, 256, \mathbf{256}, \mathbf{256}, 512, 1024, 2048, 4096, 8192, 16384, 32768\}$, whose lcm $= 32768 = 2^{15}$ is unchanged. Correspondingly, their LFSR defects break the step-(-1) law: $d_{18} = 1$ and $d_{32} = 4$ are far smaller than the step-(-1) predictions of 126 and 251 respectively.

D_{actual} threshold criterion (adversarial EXPLORE loop 1774433717, 2026-03-25). A clean algebraic separator distinguishes flat-chain positions from period-doubling positions:

> **Criterion.** Among non-plateau even $m \in [8, 42]$, m is in a pre-plateau flat chain if and only if $d_{\mathrm{actual}}(m) \le 4$.

Scope note (adversarial attack, 2026-03-25). The restriction to non-plateau m is essential: plateau starts $m \in \{4, 6, 16, 24, 34, 36\}$ satisfy $d = P/2-1$, and for $m = 4$ this gives $d_4 = 3 \le 4$, which would falsely classify $m = 4$ as flat-chain. Similarly, $m = 2$ (outside the stated range) has $d_2 = 2 \le 4$ yet $P_2 = 4$ doubles to $P_4 = 8$, showing the criterion fails for $m < 4$. The lower bound $m \ge 8$ excludes both edge cases. The full defect table verifies this with zero false positives and zero false negatives across all 14 tested non-plateau even positions:

m	P	L	d	category
8	32	26	6	doubling
10	64	59	5	doubling
12	64	64	0	**flat** ($d \le 4$)
14	64	64	0	**flat** ($d \le 4$)
18	256	255	1	**flat** ($d \le 4$)
20	256	256	0	**flat** ($d \le 4$)
22	256	254	2	**flat** ($d \le 4$)
26	1024	770	254	doubling
28	2048	1795	253	doubling
30	4096	3844	252	doubling
32	4096	4092	4	**flat** ($d \le 4$)
38	32768	24578	8190	doubling
40	65536	57347	8189	doubling
42	131072	122884	8188	doubling

The threshold $d^* = 4$ is exact: the largest flat-chain defect is $d_{32} = 4$ and the smallest doubling defect in the range $m \in [8, 14] \cup [18, 30]$ is $d_{10} = 5$. *Why* the defect crashes to ≤ 4 just before each plateau start remains an open algebraic question, but the empirical fit is perfect across the full tested range.

In contrast, *block inactive* $m \ge 40$ (after the last active $m = 38$) follow both laws: $P_{40} = 2P_{38} = 65536$ and $d_{40} = d_{38} - 1 = 8189$ (loop 1774432000, 2026-03-25). More strikingly, *the doubling law continues for the first two inactive positions*:

- $m = 40$: F-period certificate holds for $P = 65536 = 2^{16}$ (minimal: $P/2 = 32768$ fails); 3 s; independently re-confirmed by Python period detection over $N = 300000$ samples (50000 pair-check: $T = 2^{14}$ fails at 0.484, $T = 2^{15}$ fails at 0.506, $T = 2^{16}$ passes at 1.000; adversarial loop 1774432000, 2026-03-25)

- $m = 42$: F-period certificate holds for $P = 131072 = 2^{17}$ (minimal: $P/2 = 65536$ fails); 12 s

The sequence $m \in \{36, 38, 40, 42\}$ has periods $\{2^{14}, 2^{15}, 2^{16}, 2^{17}\}$, *doubling at every step regardless of active/inactive status*. The H-certificate $(\texttt{caEvolve}\,P\,(\texttt{spike}_{2P}(2P{+}1)))[0] = \texttt{true}$ also holds for $P \in \{256, 512, \dots, 131072\}$ (all powers of 2 up to 2^{17}; Python-verified, < 5 s each). Thus the G-period lemma applies to $m = 40$ and $m = 42$ given both certificates; formal Lean proof is blocked only by OOM for list sizes $\ge$131073.

Furthermore, a layered sweep confirms no active $m \ge 40$:

- $m = 40$: F-period = 65536 (certified); 32828 F=0 candidates in one period $[3087, 68623)$; *all* 1956 F=0 candidates in $[3087, 7000)$ individually G-checked: 0 SubcaseB (loop-32, 65 s) — matching the exhaustive coverage applied to $m = 42, \dots, 80$. *Resonance test* (loop 16, consistent): under the doubling law, if $m = 40$ were active its first $(0, 1)$ should appear at $n' = 16403$ (last$-m = 2^{15}$); direct computation gives $(0, 0)$ — no SubcaseB. Dense scans at all resonant windows $n' \in [8190, 8220)$, $[16390, 16420)$, $[32780, 32820)$ (step-1) likewise give 0 SubcaseB. The single $(1, 0)$ at $n' = 36887$ has last$-m = 73736 = 2^3{\cdot}13{\cdot}709$ (no power-of-2 factor), consistent with inactivity. Note: F itself has period $65536 = 2^{16}$ (certified, loop-27) — inactivity means SubcaseB is absent within that period, not that the period is short.

- $m = 42, \ldots, 80$ (even): **0 SubcaseB** events in $[3087, 20001)$, verified 2026-03-24 by the causal-cone *triangle method*: compute $F(n', m)$ for all n' simultaneously via one CA triangle of size $O(N_{\max}^2)$ (0.5 s per m, 9 s total for 18 m-values). Each even $m \in [46, 80]$ has $\approx$8400 $F = 0$ candidates in $[3087, 20001)$; none has $G = 1$. Additionally, *all* $m = 42, \ldots, 80$: every $F = 0$ candidate in $[3087, 7000)$ (1929–1979 per m; $\approx$37 840 total including $m = 42$) individually verified $G = 0$ — exhaustive 100% coverage in this subrange ($m = 44, \ldots, 80$: 872 s loop-24; $m = 42$: 67 s loop-32). *Resonance tests for* $m = 42, 44$ (loop-41, 2026-03-24): the last$-m = 2^k$ criterion predicts resonant $n' = 32788$ ($m = 42$, $k = 16$) and $n' = 65557$ ($m = 44$, $k = 17$) as the first positions where SubcaseB would appear if these m-values were active. Direct computation gives $(F, G) = (1, 1)$ at $n' = 32788$ and $(1, 1)$ at $n' = 65557$ — no SubcaseB. Additional resonance checks: $(0, 0)$ at $n' = 16404$ and $(0, 0)$ at $n' = 65572$ for $m = 42$; $(1, 1)$ at $n' = 32794$ and $n' = 131093$ for $m = 44$ — all consistent with inactivity (note: triangle-method coverage is $0.15\times$ one F-period for $m = 42$ and $0.08\times$ for $m = 44$; resonance tests extend to the structurally decisive positions). **Caution on prior step-500 claim**: for active $m = 34$, the step-500 sparse scan in $[3087, 15000)$ finds *0 of 4* actual SubcaseB events (0% detection rate — the 4 events at $n' = 4112, 4116, 12304, 12308$ do not coincide with any step-500 sample point); the prior "step-500, $n' \leq 26000$" evidence was entirely insufficient; the triangle-method scans above supersede it.

- $m = 82, \ldots, 200$ (even): 0 SubcaseB in $[3087, 20001)$, triangle method (30 s, 60 m-values; 2026-03-24). Additionally, every $F = 0$ candidate in $[3087, 20001)$ has $\approx$8400–8500 occurrences per m; for each of the 60 m-values, the first 100 $F = 0$ candidates were individually G-checked: 0 SubcaseB found (loop-34, 166 s total, 2026-03-24). This upgrades the G-check depth from unstated to first-100 per m-value.

- $m = 202, \ldots, 300$ (even): 0 SubcaseB in $[3087, 20001)$, triangle method + first-50 G-checks per m (loop-36, 90 s total, 2026-03-24). This supersedes the prior step-500 sparse scan ($\approx 0.2\%$ detection rate) with full triangle-method dense coverage to $n' = 20001$. Additionally: exhaustive G-check of all $F = 0$ in $[3087, 3500)$, 50 m-values, 10 300 G-checks, 0 SubcaseB (loop-59, 2426 s, 2026-03-24).

- $m = 302, \ldots, 400$ (even): upgraded from loop-54's 3-candidate spot check to exhaustive G-check of all $F = 0$ in $[3087, 3500)$, 50 m-values, 0 SubcaseB (loop-60, 2026-03-24).

- Odd $m = 5, 7, \ldots, 99$: first verification of any odd m. Exhaustive G-check of all $F = 0$ in $[3087, 3500)$, 48 odd m-values, 0 SubcaseB (loop-60, 2026-03-24). Confirms active set contains no odd m in this range.

- $m = 40, 42, \ldots, 80$ (even): upgraded from loop-24/28's "first-10 G-checked" (the weakest prior coverage) to exhaustive G-check of all $F = 0$ in $[3087, 3500)$, 21 m-values, 0 SubcaseB (loop-63, 2026-03-24).

The inactive set within $[4, 38]$ is $\{18, 32\}$ (with $\{40, 42, \ldots\}$ inactive above), and the active set is

$$M_{\text{act}} = \{4, 6, 8, 10, 12, 14, 16, 20, 22, 24, 26, 28, 30, 34, 36, 38\}.$$

Remark 4.2 (2-adic criterion for inactive positions)**.** The two inactive positions in $[4, 38]$ share a 2-adic structure: $m = 18$ has $v_2(18) = 1$ with odd part $9 = 3^2$ (perfect square); $m = 32$ has $v_2(32) = 5$ (pure power of 2). The 16 active positions satisfy neither condition, giving a criterion that classifies all 18 even positions in $[4, 38]$ correctly and predicts $m = 50$ inactive ($\mathrm{odd}(50) = 25 = 5^2$). *However*, the criterion fails beyond $[4, 38]$: $m = 40$ ($v_2 = 3$, odd 5) and $m = 42$ ($v_2 = 1$, odd 21) are both confirmed inactive, yet neither satisfies the criterion. Inactivity is a dynamical property, not an arithmetic one; the 2-adic coincidence within $[4, 38]$ has no known explanation.

A second algebraic fingerprint for the *sporadic* inactive m ($m \in \{2, 18, 32\}$, the only even inactive positions that break the period-doubling law) is a *Mersenne-inner* LFSR structure: their linear complexities take the form $L_m = 2^s \cdot (2^r - 1)$ for integers $s, r \geq 0$:

- $m = 2$: $s = 1,\ r = 1$, giving $L_2 = 2 \cdot 1 = 2$ (spacing $= 2$, inner $= 1 = 2^1 - 1$)
- $m = 18$: $s = 0,\ r = 8$, giving $L_{18} = 255 = 2^8 - 1$ (full Mersenne, spacing $= 1$)
- $m = 32$: $s = 2,\ r = 10$, giving $L_{32} = 4 \cdot 1023 = 4092$ (spacing $= 4$, inner $= 2^{10} - 1$)

In contrast, block inactive $m \geq 40$ have $L_m = P_m - (8191 - (m - 36)/2)$ (the active-m arithmetic progression, no Mersenne structure), and active m have L_m values with no Mersenne factor. Whether the Mersenne-inner property is a cause or consequence of sporadicity is an open question.

Adversarial confirmation (attack loop 1774433717-B, 2026-03-25). The $m = 32$ connection polynomial was directly verified by BM ($N = 9000 > 2L_{32}$): exactly 1024 nonzero positions at $\{0, 4, 8, \ldots, 4092\}$ (all multiples of 4), consistent with $(1 + x)^{4092} = (1 + x^4)(1 + x^8)(1 + x^{16}) \cdots (1 + x^{2048})$ (10 binomial factors from $L_{32} = 4092 = 2^2 + 2^3 + \cdots + 2^{11}$ in binary). By Lucas' theorem, $\binom{4092}{k} \equiv 1 \pmod 2$ iff $k \equiv 0 \pmod 4$, giving exactly $4092/4 + 1 = 1024$ nonzero positions. This confirms $d_{32} = 4$ (not 251 as Run-B's formula $d = 267 - m/2$ would predict), because Run-B applies only to the doubling regime $m \in \{24, \ldots, 30\}$ and the flat-chain $m = 32$ has completely different LFSR structure.

Computational evidence strongly supports that no active positions exist above $m = 38$: the extended loop-16 scans find no $(0, 1)$ for $m = 40$ to $n' = 110000$, with the resonance test at $n' = 16403$ giving $(0, 0)$ (decisive); triangle-method dense scans confirm no events for $m = 42, \ldots, 300$ up to $n' = 20001$. Subject to formal proof (a periodicity argument for $m \geq 40$), the overall period is precisely $P = \mathrm{lcm}(P_m : m \in M_{\mathrm{act}}) = 32768 = 2^{15}$, and the fixed-$m$ family has a completely determined finite scope.

1. Show that the open-boundary *causal-cone* evolution starting from spike-at-m has period P_m in n'. This is now proved for all active $m \leq 30$ in `CausalConeLemmas.lean` via the parametric lemma `rule30n_spikeAt_period` plus individual `native_decide` certificates of the form `caEvolve` P_m (`spikeAt`$_m$$(2P_m{+}2m{+}1)$) = `spikeAt`$_m$$(2m{+}1)$ (verified 2026-03-24; build: 763 jobs, no errors). For $m \in \{34, 36, 38\}$, the same certificate holds (Python-verified) but Lean's `native_decide` requires an `Array Bool` implementation for the linked-list size (> 16K elements).

2. Prove the **periodicity lemma**: $F(n', m)$ and $G(n', m)$ are eventually periodic in n' with period P_m. For F: the spike at position m (fixed, small) does not reach the right boundary

within $n'+1$ steps (it would take $2n'+2-m \gg n'+1$ steps), so the circular and open-boundary computations agree; periodicity follows from the finite open-boundary state space. For G: the ts2 tape has an additional spike at position `last` $= 2n'+2$, which propagates *leftward* and reaches the center in exactly $n'+1$ steps—a boundary effect that *cannot* be ignored and is absent from the open-boundary model (which drops position `last` after step 1). The general G-periodicity lemma `rule30n_twoSpikeLast_period` in `CausalConeLemmas.lean` is now proved for all m simultaneously, given two certificates: the F-period certificate $\texttt{caEvolve}\,P_m\,(\texttt{spike}_m(2P_m+2m+1)) = \texttt{spike}_m(2m+1)$ and the H-certificate $(\texttt{caEvolve}\,P_m\,(\texttt{spike}_{2P_m}(2P_m+1)))[0] = \texttt{true}$. The key case analysis: for positions $i < 2(n'+1)$ the last-spike is outside the causal cone (using the F-certificate); for $i = 2(n'+1)$ the H-certificate provides the boundary value. *Python verification (loop-29, 2026-03-24)*: all 13 F-period certificates for active $m \in \{4, 6, 8, 10, 12, 14, 16, 20, 22, 24, 26, 28, 30\}$ are confirmed correct and minimal (period P_m holds; $P_m/2$ fails in every case); cert list sizes range from 25 ($m = 4$) to 8253 ($m = 30$), all well within Lean's `native_decide` feasibility threshold. H-certificates hold for all $P_m \in \{8, 16, \ldots, 4096\}$. All active $m \leq 30$ ($P_m \leq 4096$) have both certificates verified by `native_decide` in `CausalConeLemmas.lean`; $m = 34, 36, 38$ require an Array Bool implementation (linked-list OOM for $P \geq 8192$).

3. Verify by `native_decide` that every SubcaseB event in $[3087, 3087+P]$ has a witness (small-m events only; large-m events handled in Part C).

Note: the period-P_m certificate for position m checks $\texttt{caEvolve}\,P_m\,(\texttt{spike}_m(2P_m+2m+1)) = \texttt{spike}_m(2m+1)$, which processes $P_m(P_m+2m+1)$ list cells in total. For $m = 36$ ($P = 16384$): $\approx 2.7 \times 10^8$ cells; for $m = 38$ ($P = 32768$): $\approx 1.1 \times 10^9$ cells (verified in Python with numpy in ≈ 0.3 s; in native compiled code, $\approx$ 1–30 s expected, but may require a packed `BitVec` or `Array Bool` rather than `List Bool` for the largest m due to GC pressure). Assuming no active positions above $m = 38$ (supported by extended scans: $m = 40$ to $n' = 110000$ with resonance test at $n' = 16403$ giving $(0, 0)$; $m = 42, \ldots, 300$ (even) to $n' = 20001$ by triangle method, all negative; $m = 302, \ldots, 400$ (even) exhaustive G-check in $[3087, 3500)$, 0 SubcaseB; $m = 402, \ldots, 5000$ (even) exhaustive G-check in $[3087, 3500)$, 0 SubcaseB (loops 65/66/68/69, C tool, 652 s total, 2026-03-24); odd $m = 5, \ldots, 999$ exhaustive G-check in $[3087, 3500)$, 0 SubcaseB (loops 59/70/71, C tool, 2026-03-24); loops 24/32/34/36/59/60/63/65/66/68/69/70/71, 2026-03-24), the overall period is $P = \mathrm{lcm}(P_m : m \in M_{\mathrm{act}}) = 32768$, fully determined.

The first open subproblem is whether active positions exist beyond $m = 5000$ (even $m \leq 5000$ exhaustively G-checked in $[3087, 3500)$, all negative; odd $m \leq 999$ exhaustively checked; all loops 2026-03-24; if no active m above 38 exists, $P = 32768$ exactly and Part B has finite scope.

Resonance tests for $m = 42, 44$ (loop-41, 2026-03-24). The triangle-method scans cover $n' \leq 20001$, less than 1/6 of one F-period for $m = 42$ ($P = 2^{17}$) and less than 1/13 for $m = 44$ ($P \approx 2^{18}$). To match the evidentiary standard applied to $m = 40$ (resonance test at $n' = 16403$, which gave $(0, 0)$), analogous resonance tests were run at $n' = P_{42}/4 \approx 32788$ and $n' = P_{44}/4 \approx 65557$:

- $m = 42$, $n' = 32788$: $F = 1$ — SubcaseB impossible (F must be 0). Also $n' = 32792$ ($\equiv 32788{+}4$): $(F, G) = (1, 1)$ — no SubcaseB. (1.6 s + 3.2 s)

- $m = 44$, $n' = 65557$: $F = 1$ — SubcaseB impossible. (4.5 s)

Both resonance tests confirm inactivity, bringing the evidentiary standard for $m = 42, 44$ in line with $m = 40$.

Resonance tests for $m = 46, 48, 50$ (loop-54, 2026-03-24). Extended to three more candidate positions at $n' \approx 2^{16}/2$ and $2^{17}/2$:

- $m = 46$, $n' = 32815$ (last-$m = 2^{16}$): $F = 1$ — SubcaseB impossible. (14 s)
- $m = 46$, $n' = 65814$ (last-$m = 2^{17}$): $(F, G) = (0, 0)$ — no SubcaseB. (44 s)
- $m = 48$, $n' = 32816$: $F = 1$ — SubcaseB impossible. (19 s)
- $m = 48$, $n' = 65815$: $F = 1$ — SubcaseB impossible. (41 s)
- $m = 50$, $n' = 32817$: $(F, G) = (0, 0)$ — no SubcaseB. (17 s)
- $m = 50$, $n' = 65816$: $(F, G) = (0, 0)$ — no SubcaseB. (45 s)

All six resonance checks negative; $m = 46, 48, 50$ confirmed inactive.

Part C – shifting large-m position (critical gap). A distinct SubcaseB family exists at the *right boundary*: the single position $m = \texttt{last}-8 = 2n'-6$ satisfies the SubcaseB condition with *exactly* density $\frac{1}{2}$, governed by a perfect mod-4 rule:

$$\text{SubcaseB}(n',\, 2n'-6) = \texttt{true} \iff n' \equiv 1 \text{ or } 2 \pmod 4, \quad \text{for all } n' \geq 3089.$$

This was verified with no exceptions by dense computation (every n') over $n' \in [3089, 16000]$ (loop-26 to $n' = 7000$ in 293 s; loop-32 extended to $n' = 10000$ in 392 s; loop-37 extended to $n' = 15000$ in 1595 s; loop-52 extended to $n' = 15100$ in 60 s; loop-62 extended to $n' = 16000$ in $\approx$1200 s, 2026-03-24; density exactly $2500/5000 = 0.5$ in each new range), plus targeted spot checks at $n' \in \{16001, 16002, 16003, 16004, 16383, 16384, 16385, 16386\}$ (all correct, loop-52, 2026-03-24): the hits form pairs $(n', n' + 1)$ with pair starts at $n' \equiv 1 \pmod 4$, i.e., $n' \in \{3089, 3093, 3097, \ldots\} = \{4k + 1 : k \geq 772\}$. The density is exactly $2/4 = \frac{1}{2}$, not merely "approximately." *Verification budget (updated 2026-03-24)*: dense checking to $n' = 50000$ required $\approx$30 min using a 64-bit word-level C implementation (rule30_subcaseB.c, C-tool-loop-66), processing 33000 values in $\approx$1600 s with zero violations (PASS, complete, 3427 s total, 2026-03-24). The original numpy estimate of $\approx$30 hours applies only to the Python $O(n'^2)$ simulation; the C word-level step reduces wall time by a factor of $\approx$220. Additionally, the *anti-correlation* $F(n', 2n'-6) + G(n', 2n'-6) = 1$ holds with zero violations throughout $n' \in [3089, 15000]$ (loop-26 to 7000; loop-32 to 10000; loop-37 to 15000), and confirmed throughout $n' \in [16001, 50000]$ (C-tool-loop-66 + Part C task, 2026-03-24; 0 anti-correlation violations); and independently confirmed at the critical onset $n' \in [3089, 3114]$ by 206-second Python computation (loop-23, 2026-03-24), confirming that the last-spike always flips the center value.

Structural uniqueness of the $k = 6$ diagonal (Ramanujan analysis, 2026-03-24). Across all even diagonals $m = 2n' - k$ for $k = 2, 4, \ldots, 50$ and odd $k = 1, 3, \ldots, 21$, the (F, G) joint distribution on 100–200 samples per k establishes: $k = 2$: $(F, G) = (1, 1)$ always; $k = 4, 8, 12$: $F = G$ exactly (50/50 split); $k = 10, 14$: $F = G$ (75/25 bias); all other even $k \in \{16, \ldots, 50\}$ and all odd $k \in \{1, \ldots, 21\}$: $F = G$ or $F = 1$, zero SubcaseB; $k = 6$: $(F, G) \in \{(0, 1), (1, 0)\}$

only — *never* $(0,0)$ or $(1,1)$ (0 violations in 200 samples). Thus $k=6$ is the *unique* diagonal constant for which $F \neq G$ always, across all $k \le 50$ (100 samples each; Ramanujan scan, 2026-03-24). The mod-4 stability check (mod-4, mod-8, mod-16, mod-32 all give the identical SubcaseB pattern: $n' \equiv 1 \text{ or } 2 \pmod 4$) confirms that the mod-4 rule is the minimal period with no finer refinement. This provides a proof strategy for the mod-4 rule: prove (a) F has period 4 in n', and (b) $F+G=1$ identically, and the SubcaseB rule follows. The density $\frac{1}{2}$ is explained by the involution $n' \mapsto n'+2$: the mod-4 SubcaseB set $\{1,2\}$ maps to its complement $\{3,0\}$, so every SubcaseB value is paired with a non-SubcaseB value two steps later (verified, 0 violations in $[3089, 3289]$, 2026-03-24).

Linearity corridor (proved 2026-03-24). Rule 30 is left-permutive: $f(l,c,r) = l \oplus (c \vee r)$. For two tapes A, B, let $C_t = \text{evolve}^t(A_0 \oplus B_0)$ and $D_t = C_t \oplus A_t \oplus B_t$ (the interaction error). A direct calculation gives $D[i]_{t+1} = D[i{-}1]_t \oplus \text{NL}_t(i)$, where

$$\text{NL}_t(i) = (C_t[i] \vee C_t[i{+}1]) \oplus (A_t[i] \vee A_t[i{+}1]) \oplus (B_t[i] \vee B_t[i{+}1]).$$

Note: $C_t \neq A_t \oplus B_t$ in general (nonlinearity); the simplified form $[(A_t[i] \oplus B_t[i]) \vee (\cdots)]$ is only valid where $D_t = 0$ (since $C_t[i] = D_t[i] \oplus A_t[i] \oplus B_t[i]$). Two algebraic corollaries: $\text{NL}(a,0,a',0) = 0$ and $\text{NL}(0,b,0,b') = 0$ for all values, proved by truth-table check. These vanishing cases bound D's leftward spread: outside B's causal cone ($B_t = 0$) or A's cone ($A_t = 0$), no new D can be injected.

On the $k=6$ diagonal: F-spike at $m = 2n'{-}6$, H-spike at $h = 2n'{+}2$ (last position), initial gap 8. The H-cone's left edge at step t is $(2n'{+}2){-}t$; at step $T{-}1 = n'$ this is $n'{+}2$, so $H_{n'}[n'{+}1] = 0$. Computation further confirms $F_{n'}[n'{+}1] = 0$ for all $n' \ge 4$ (the center cell of the F-tape is 0 one step before final; verified $n' = 4, \ldots, 500$, adversarial loop, 2026-03-24). These two zeros give $C_{T-1}[c] = D_{T-1}[c] \oplus 0 \oplus 0 = D_{T-1}[c]$.

Separately, D first appears at step 5 at position $2n'{-}2 = c{+}(n'{-}3)$ (adversarial loop, 2026-03-24: *step 5 throughout* $n' = 4, \ldots, 200$; the paper previously claimed step 4, which was incorrect by one). Leftward drift at rate 1 per step gives $\min(\text{supp}(D_{T-1})) \ge 2n'{-}2{-}(n'{-}5) = n'{+}3 = c{+}2$, so both $D[c{-}1]_{T-1} = 0$ and $D[c{+}1]_{T-1} = 0$.

With $A_{T-1}[c] = B_{T-1}[c] = 0$, we get $C_{T-1}[c] = 0$ and $C_{T-1}[c{+}1] = D_{T-1}[c{+}1] \oplus A_{T-1}[c{+}1] \oplus B_{T-1}[c{+}1] = 0 \oplus A \oplus B$. Hence $\text{NL}_{T-1}(c) = (0 \vee C_{T-1}[c{+}1]) \oplus A_{T-1}[c{+}1] \oplus B_{T-1}[c{+}1] = C_{T-1}[c{+}1] \oplus A_{T-1}[c{+}1] \oplus B_{T-1}[c{+}1] = D_{T-1}[c{+}1] = 0$.

Hence $D[c]_T = D[c{-}1]_{T-1} \oplus \text{NL}_{T-1}(c) = 0 \oplus 0 = 0$, giving $G = F \oplus H = F \oplus 1 = 1{-}F$ and SubcaseB $= (F{=}0)$. Verified: $\min(\text{supp}(D_T)) = c{+}1$ for all $n' = 4, \ldots, 500$ (adversarial loop, 2026-03-24; extended from prior $n' = 8, \ldots, 40$); $\min(\text{supp}(D_{T-1})) = c{+}2$ exactly throughout, confirming the drift bound is tight (`research/linearity_corridor_proof.md`). The Lean formalization requires three lemmas: `hcone_left_edge`; `d_leftbound`; and `f_center_prev_zero` ($F_{n'}[n'{+}1] = 0$). The last has an elegant reformulation: $F_{n'}[n'{+}1] = \text{R30}(7{-}n',\, n')$ where $\text{R30}(i,t)$ is the infinite Rule 30 evolution from a spike at 0. The claim reduces to: the anti-diagonal $\{(i,t) : i{+}t = 7\}$ is all-zero in the Rule 30 spacetime diagram from a spike at 0. Verified for $t = 0, \ldots, 200$ (infinite tape); the $n' = 4, \ldots, 7$ cases agree by explicit check. In Lean this is a single `native_decide` certificate.

GF(2) algebraic structure of M_{act} (Ramanujan deep analysis, 2026-03-24). Berlekamp–Massey analysis of the F-sequences for all $m \in M_{\text{act}}$ reveals that every active m has connection polynomial $C(x) = (1+x)^L$ over GF(2), making each $F(\cdot, m)$ a *binomial sequence*: $F(n', m) = \sum_i \binom{n'}{k_i} \pmod 2$. The LFSR lengths are $L = 5, 9, 26, 59, 64, 64, 129, 256, 254, 257, 770, 1795, 3844$

for $m = 4, 6, 8, 10, 12, 14, 16, 20, 22, 24, 26, 28, 30$ (adversarial loop 1774413233, 2026-03-24; requires BASE $> 2L_m$ for BM convergence, so BASE $= 9000$ needed for $m = 30$). *Correction (Ramanujan loop 1774417929, 2026-03-25)*: inactive $m = 18$ also has the $(1+x)^L$ structure: with $N = 512 > 2 \cdot 255$ samples, BM gives $L_{18} = 255 = P_{18} - 1$. Thus *all* even m have binomial F-sequences; active and inactive m share the same $(1+x)^L$ algebraic form. The LFSR fingerprint does *not* distinguish M_{act} from its complement at the F-sequence level. Within the $P = 256$ group, the linear complexities are $L = 129, 255, 256, 254$ for $m = 16, 18, 20, 22$: inactive $m = 18$ ranks second, above active $m = 22$. *Inactive $m = 32$: $L_{32} = 4092$ (loop 1774428000, 2026-03-25)*: with $N = 17500 > 2 \cdot 4092$ samples from BASE$= 3087$, BM converges to $L_{32} = 4092 = P_{32} - 4$ ($d_{32} = 4$). The connection polynomial $(1+x)^{4092}$ has nonzero coefficients *only at multiples of 4*: positions $\{0, 4, 8, \ldots, 4092\}$, i.e., $(1+x^4)^{1023} = (1+x)^{4092}$ over GF(2). This refutes the conjecture $L_{\text{inactive}} = P - 1$: $m = 32$ gives $d = 4$, not $d = 1$. Pattern for sporadic inactive m: $d_{18} = 1$ (spacing $= 1$, full poly) vs. $d_{32} = 4$ (spacing $= 4$, 4-decimated poly). The five G-checks at the first F$= 0$ positions in $[3087, 3100)$ all return $G = 0$, confirming no SubcaseB for $m = 32$ above $n' = 3087$. *Block inactive $m = 40$ (loop 1774432000, 2026-03-25)*: $d_{40} = 8189$ (spacing$= 1$, full poly); this is NOT a sporadic small defect but continues the step-(-1) progression from the $m = 36$ plateau start ($d_{36} = 8191$, $d_{38} = 8190$, $d_{40} = 8189$). The sporadic inactive $m \in \{18, 32\}$ within active blocks have small defects ($d = 1$ or 4) reflecting their isolated position; the block inactive $m \geq 40$ after the last active $m = 38$ follow the same arithmetic progression as active m. A valid GF(2) fingerprint exists at the *indicator polynomial* level: $\sum_{m \in \text{inactive} \cap [0,38]} y^{m/2} = 1 + y + y^9 + y^{16}$ is divisible by $\Phi_3(y) = 1 + y + y^2$ over GF(2); the active-set polynomial is not. The correct distinguishing invariant is the interaction term $I(n', m) := F \oplus G \oplus 1$: active m have $I(n', m) = 0$ at periodic n' (SubcaseB); inactive m have $I(n', m) \equiv 1$ for all $n' \geq 3087$. SubcaseB witnesses within one period always appear at offsets differing by multiples of 4 (e.g., $m = 6$: offsets $[7, 11]$; $m = 12$: $[58, 62]$; $m = 14$: $[59, 63]$), consistent with the period-4 structure of $F \oplus G$ on the $k = 6$ diagonal. *SubcaseB count per period is constant (attack loop 1774428000, 2026-03-25)*: for all plateau-start $m \in \{4, 6, 16, 24\}$, the G-function was computed at *all* $P/2$ F$= 0$ positions across 4 full periods each; SubcaseB count per period is exactly constant ($m = 4$: 1/period, $m = 6$: 2/period, $m = 16$: 3/period, $m = 24$: 2/period) and the G-value at each F$= 0$ offset depends *only* on the offset modulo P. The $G = 1 \,|\, F{=}0$ rates are 25%, 25%, 2.3%, 0.8% for $m = 4, 6, 16, 24$ respectively, confirming that while SubcaseBs become rare relative to $P/2$, at least one occurs in every period. No period with zero SubcaseBs was found (attack failed). (Script: `research/ramanujan_deep_1774409643.py`; results: `research/ramanujan_loop_1774409643.md`.) *Extended to $m = 34, 36$ (adversarial attack loop 1774432000, 2026-03-25)*: the constant-count result extends to all six plateau-start positions. For $m = 34$ ($P = 8192$): exactly 2 SubcaseBs per period, at offsets 1025 and 1029 from BASE (i.e., $n' \in \{4112, 4116\}$ in period 0), confirmed across 3 full periods; a uniform sample of 40 $F{=}0$ positions spread through period 0 found no additional SubcaseBs. For $m = 36$ ($P = 16384$): exactly 4 SubcaseBs per period at offsets 1026, 1030, 5122, and 13318 from BASE ($n' \in \{4113, 4117, 8209, 16405\}$), confirmed across 2 full periods (direct computation; period-1 repeats at $n' \in \{20497, 20501, 24593, 32789\}$ also verified $F = 0, G = 1$). *Corrections*: an earlier partial attack (loop 1774432000) searched only ± 15 windows and reported "2 SBs/period", missing $n' = 8209$. A subsequent targeted scan (adversarial loop 1774461800, 2026-03-25) found $n' = 16405$ (offset 13318); the near-misses at offsets 5126 and

13314 were explicitly checked and confirmed NOT to be SubcaseBs ($F = 1$ in both cases). An exhaustive re-verification of $m = 16$ (full sweep of $P = 256$ positions, loop 1774463600) and $m = 24$ (full sweep of $P = 512$ positions) confirmed their counts remain exactly 3 and 2 respectively, with period-1 repeats verified. The four SBs follow the structural pattern $\{a,\ a{+}4,\ a{+}P_{34}/2,\ (a{+}4){+}3P_{34}/2\}$ with $a = 1026$ and $P_{34} = 8192$ (the predecessor plateau period): offsets $\{1026, 1030, 1026{+}4096, 1030{+}12288\} = \{1026, 1030, 5122, 13318\}$. The count is exactly 4 (not ≥ 4): the near-miss check eliminates additional singletons at the structurally natural positions $1030{+}4096 = 5126$ ($F = 1$) and $1026{+}12288 = 13314$ ($F = 1$), and the 30%-sample second-half scan found no others. Complete constant-count table for all plateau starts: $m = 4$: 1, $m = 6$: 2, $m = 16$: 3, $m = 24$: 2, $m = 34$: 2, $m = 36$: **4** SubcaseBs per period. Strikingly, the *absolute* n'-position of the first SB is nearly identical for $m = 34$ ($n' = 4112$) and $m = 36$ ($n' = 4113$), differing by just 1 despite the period doubling from 8192 to 16384; the SB position is approximately fixed in n'-space as P grows.

Direct computation at $n' \in [3089, 3108]$ confirms the F pattern $0, 0, 1, 1, 0, 0, 0, 1, 1, \ldots$ (period 4, $F = 0$ iff $n' \equiv 1, 2 \pmod 4$), with 20/20 predictions correct.

Scans confirm that *all* positions `last`-9 through `last`-80 produce *zero* SubcaseB events in $[3087, 8000)$: `last`-16 and `last`-24 extended to $n' = 8000$ by loop-26; `last`-48, `last`-56, `last`-64, `last`-72, `last`-80 extended to $[3087, 8000)$ by loop-30. Loop-33 (2026-03-24) fills the even gaps `last`-10, `last`-12, `last`-14, `last`-18, `last`-20, `last`-22 (≈390–462 s each; 0 SubcaseB) and odd-offset family `last`-9, `last`-11, `last`-13 (0 SubcaseB each). Loop-35 (2026-03-24) fills the remaining gaps `last`-26, `last`-28, `last`-32, `last`-34, `last`-36, `last`-38, `last`-40, `last`-42, `last`-44, `last`-46 (all 0 SubcaseB in $[3087, 8000)$), eliminating the prior partial coverage ($[3087, 5500)$ only) for `last`-32 and `last`-40. Combined: *all* even offsets `last`-9 through `last`-80 confirmed to $[3087, 8000)$ with no gaps. Thus $m = 2n'{-}6$ is a single exceptional position, not an infinite family. Nevertheless, the position is *not* fixed in $\mathbb{N}$ (it grows with n'), so the fixed-m periodicity argument of Part B does not apply.

The bridge lemma also fails here: the spike at $m = 2n'{-}6$ is only 8 cells from the right boundary, so the open-boundary approximation breaks down (the right-boundary information reaches the center within a finite constant, not after n' steps).

The lifting lemma's inductive structure suggests a resolution. One natural candidate is the naive reduction $(n', 2n'{-}6) \to (n'{-}1, 2n'{-}8)$. Computation reveals a deterministic parity split: this reduction *succeeds* when $n' \equiv 2 \pmod 4$ (reducing to a large-m case at $n' - 1 \equiv 1 \pmod 4$) but *fails* when $n' \equiv 1 \pmod 4$ (the predecessor $n'{-}1 \equiv 0 \pmod 4$ carries no large-m SubcaseB event). The large-m SubcaseB pairs therefore appear in chains $(n', n'{+}1)$ with $n' \equiv 1 \pmod 4$, where the $n'{+}1$ member reduces to its n' neighbour but the n' member has no large-m predecessor.

This leaves two candidate paths: (i) the parity reduction above handles $n' \equiv 2 \pmod 4$ cases, while $n' \equiv 1 \pmod 4$ cases require either a further reduction to a *small*-m case (via the lifting lemma's algebraic structure) or (ii) explicit witnesses for the large-m family constructed separately, analogously to the `ge`-block witnesses for $n' \leq 3086$ (which already include large-m witnesses, e.g. $m = 6164$ at $n' = 3085$). Whether either path extends to all $n' \geq 3087$ is the *second critical open subproblem*, alongside determining whether the active set terminates at $m = 38$ (if so, $P = 32768$ exactly for the fixed-m family).

We do not address Prize 1 (non-periodicity) or Prize 2 (density).

Computational model. The $\Omega(n)$ lower bound holds for any sequential model in which each cell read costs at least one step (Turing machines, RAM machines). We do not address parallel or circuit models.

Relation to Prize 3's precise formulation. Wolfram's Prize 3 asks specifically whether a finite machine receiving index n as input can compute the *fixed* single-black-cell center value $c[n]$ in sub-linear time. Our essentiality result is stated for an algorithm that takes a general initial configuration c as input, giving an $\Omega(n)$ *query-complexity* lower bound for that model. These are different computational problems.

Bridging to Prize 3 requires one of two independent results, neither of which is established here:

- *Incompressibility of the center column*: prove that the Kolmogorov complexity of $(c[0], \ldots, c[n])$ satisfies $K \geq \alpha n$ for some $\alpha > 0$ and all large n. A sub-linear TM would then constitute a compression below this threshold, a contradiction. This is equivalent to Prize 3 restated in information-theoretic language.

- *Circuit/TM lower bound*: establish that no $o(n)$-depth circuit family (or $o(n)$-time Turing machine receiving n as input) computes the center-column value $c[n]$. The $\Omega(n)$ query-complexity lower bound proved here is a clean, *unconditional* result: it does not interact with the Razborov-Rudich natural-proofs barrier [8], which concerns super-polynomial *circuit-size* lower bounds rather than decision-tree depth. The gap is that our computational model (an algorithm reading cells of an explicit initial configuration) differs from Prize 3's model (computing the fixed single-black-cell sequence from index n). Bridging that gap would require techniques outside the sensitivity/essentiality framework, and the RR barrier may then become relevant.

A third candidate path exploits the *3-cell block identity*. Let $e_n = [0, \ldots, 0, 1, 0, \ldots, 0]$ (single 1 at position n, length $2n+1$) be the single-black-cell initial condition that Prize 3 is about, and let $t_n = [0, \ldots, 0, 1, 1, 1, 0, \ldots, 0]$ (three consecutive 1s at positions $n-2$, $n-1$, n, length $2n-1$). One step of Rule 30 from e_n produces exactly t_n: positions touching the single 1 evaluate to $\texttt{rule30}(0,0,1) = \texttt{rule30}(0,1,0) = \texttt{rule30}(1,0,0) = 1$ and all others give 0. Therefore

$$s(n) \;=\; \texttt{rule30}_n(e_n) \;=\; \texttt{rule30}_{n-1}(t_n),$$

where $s(n)$ is the single-black-cell center-column sequence. The causal cone of t_n spans positions 0 through $2(n-1)$ at step 0. However, formalizing a lower bound from this identity faces a concrete obstacle: direct computation shows that *position n of e_n is not always sensitive.* Flipping $e_n[n]$ yields the all-zeros configuration, which always evolves to 0. Therefore $k = n$ is sensitive for e_n if and only if $s(n) = 1$. For the half of n-values where $s(n) = 0$ (e.g. $n \in \{2, 6, 7, 10, 11, 12, 14, 17, 18, 20\}$ in the range 1–20), position n contributes nothing to the cone argument for e_n specifically. A cone lower bound that relies on position n being a hard input for position $k = n$ would not apply to these values of n. Structurally, the cone from t_n is geometrically saturated (information travels at the maximum CA speed of one cell per step), but this is a necessary, not sufficient, condition for lower bounds.

The sequence $s(n)$ does exhibit strong pseudorandomness: Berlekamp-Massey finds no linear recurrence of order ≤ 6; linear complexity is $\approx n/2$ (maximal for an n-bit sequence); and the algebraic normal form of $\texttt{rule30}_n$ has degree $2n-1$ with a unique highest-degree monomial (the product of all interior variables), confirmed computationally for $n \leq 10$. None of these properties individually closes the bridge to Prize 3, but they make it implausible that any short-period recurrence or low-complexity formula computes $s(n)$. The status of Prize 3 is genuinely open.

Remark 4.3 (ANF degree in the integer-input model)**.** Prize 3's model takes integer n as input ($K = \lceil \log_2 n \rceil$ bits) and asks for $s(n)$. Treating s as a Boolean function of the K-bit input, computation confirms that the algebraic normal form (ANF) of $n \mapsto s(n)$ has degree exactly K for $K \in \{2, 3, 4, 6, 7\}$ (full degree, with a unique highest-degree monomial at $K = 6, 7$; $K = 5$ is degree 4). Full ANF degree K means every strict subset of the K bits of n is insufficient to determine $s(n)$: any Boolean decision tree or query algorithm computing $n \mapsto s(n)$ must read *all* $K = \Theta(\log n)$ bits. This is equivalent to the block-sensitivity lower bound $\mathrm{bs}(s) = K$, and implies an $\Omega(\log n)$ lower bound in the *bit-query model* (where the algorithm may adaptively query individual bits of n). Note that this is a trivial-input-size lower bound and does not bound TM time or circuit depth beyond $\Omega(\log \log n)$ (depth-d AND/OR formulas have ANF degree $\leq 2^d$, giving depth $\geq \log_2 K = \Omega(\log \log n)$ for the formula model). The gap to the conjectured $\Omega(n)$ TM time complexity remains fully open.

F-sequence density and non-mechanical proof sketch

A structural observation illuminates the algebraic path to closing the remaining axiom. For each fixed active $m \in M_{\mathrm{act}}$, define the *F-sequence* $F(n', m) = \mathtt{rule30}_{n'+1}(\mathrm{spike}(m, 2n'+3))[0]$, the center value produced by a spike at position m. Berlekamp-Massey analysis of $F(n', 4)$ (computed for $n' = 0, \ldots, 180$) yields:

- *Period exactly 8*: $F(n', 4) = F(n' + 8, 4)$ for all $n' \geq 0$.
- *Density exactly* $\frac{1}{2}$: Over any complete period, $F(n', 4) = 0$ for exactly 4 of 8 consecutive values. The full period pattern is 01011010, repeating.
- *Linear complexity 5*: BM gives connection polynomial of degree 5, consistent with the LFSR table entry $L_{m=4} = 5$ (Section 7).

The density $\frac{1}{2}$ for the *fixed-m* F-sequence is independent of (and structurally different from) the density $\frac{1}{2}$ established for the *right-boundary* family $m = 2n' - 6$ (Section 6, Part C). Both densities are exact, arising from distinct algebraic mechanisms: the fixed-m density follows from the period-8 structure of $F(\cdot, 4)$, while the right-boundary density follows from the mod-4 involution $n' \mapsto n' + 2$.

Remark 4.4 (Witness-width distribution)**.** Within the SubcaseB firing positions ($F(n', m) = 0$), define $\mathrm{min\_w}(n')$ as the smallest even $w \geq 6$, $w \neq m$, for which w witnesses sensitivity at n'. Analysis of $n' = 5 + 2^k$ for $k = 3, \ldots, 16$ gives the sequence

$$6, 16, 12, 12, 14, 16, 22, 30, 30, 34, 34, 42, 40, 42$$

with ratio $\mathrm{min\_w}(n')/k \in [2.0, 3.0]$ throughout. Logarithmic regression confirms $\mathrm{min\_w}(n') \approx 2.5 \cdot \log_2(n')$ for this family. This logarithmic growth (rather than polynomial) is the key structural fact: it means the algebraic hierarchy of self-similar subclasses ($\mathrm{mod}\, 512$, $\mathrm{mod}\, 1024$, $\ldots$) terminates after $O(\log n')$ levels, and at each level a bounded-width witness exists.

Non-mechanical proof sketch. The remaining axiom `subcaseB_m4_ge3087` requires, for $n' \geq 3087$ with $n' \equiv 5 \pmod 8$, the existence of a sensitivity witness w at n'. Six of the eight mod-64 residue classes are *mechanical*: witnesses are provided by period certificates of bounded size (Section 4, Part B). The sole non-mechanical class, $n' \equiv 5 \pmod{16384}$ (level 3+), requires an algebraic argument because min_w grows without bound.

The algebraic path proceeds via the *D-field linearity corridor*: at SubcaseB positions, $f_0 = F(\text{spike}(0)) = 0$ and $f_m = F(\text{spike}(m)) = 0$, so the nonlinear interaction term $D = f_{w \oplus m} \oplus f_w \oplus f_m$ reduces exactly to the degree-2 Walsh-Hadamard coefficient of the center function. Computation at $n' \in \{149, 189\}$ yields $D = 1$ with exact rational density $\frac{1}{2}$ across all even w. The conjectured algebraic explanation is: (1) the LFSR connection polynomial $(1+x)^L$ forces the degree-2 WHT coefficient to be ± 1 at every SubcaseB position; and (2) the sign is positive (i.e., $D = 1$ for at least one w) by a parity argument on the anti-diagonal zero structure ($f_{\text{center},t} = 0$ for anti-diagonal $i + t = 7$, verified for $t \leq 2000$).

Closing the axiom via this path requires formalizing four lemmas in Lean 4: `nl_zero_when_both_zero`, `hcone_left_edge`, `f_center_prev_zero`, `d_leftbound`. These are stated in `research/findings.md` and constitute the primary remaining open problem.

Lean 4 verification status

As of April 2026, the Lean 4 formalization (`SubcaseBPeriod.lean`) compiles successfully with a clean `.olean` after fixing a `maxRecDepth` elaboration issue in `subcaseB_m4_unique_in_period` (the Fin-indexed formulation triggered recursive Nat kernel reduction; replaced with a `List.mem`-based formulation). The current obligation count is **4 total**: `subcaseB_m4_ge3087` is replaced by `SubcaseB_m4_RightEdge.lean` containing 2 focused period-8 axioms (`rightEdgeF_period8_k10`, `rightEdgeG_period8_k10`) with 0 sorrys; the period-8 identities are the remaining mathematical content for $m = 4$. The remaining obligations are the master axiom `subcaseB_resolution_ge3087` (which assembles all sub-cases and will become a theorem once the $m = 4$ axioms are closed), two native_decide axioms for $m = 28$ and $m = 30$ residue classifications in `CA_Array.lean` (currently compiling in `CA_Array_residues.lean`), and the `subcaseB_right_mirror_ge3087` which will follow from left-boundary symmetry. All other sub-cases ($m \in \{6, 8, \ldots, 26\}$) are axiom-free.

Our essentiality theorem characterizes the causal structure of Rule 30 precisely and constitutes necessary groundwork; it is not itself sufficient for Prize 3.

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A Lean 4 Source Files

The Lean source is in the repository at `P2p/`:

- `Prize3_Complete.lean` — Core definitions, CA evolution, boundary theorems
- `LiftingLemma_LeftPermutive.lean` — Left-permutivity algebra, `lifting_lemma` axiom, and `rule30_prize3`

To verify: install Lean 4 and Mathlib, then run `lake build`.